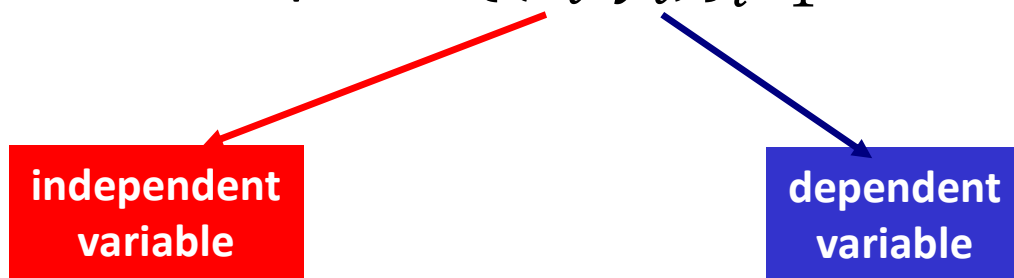


Agenda

- Short review
- Inferences on the mean of Y
- Inferences on the prediction of Y
- Variability explained by the LR model
- Validating the assumptions

Simple linear regression (1)

- The sample: $\{(x_i, y_i)\}_{i=1}^n$



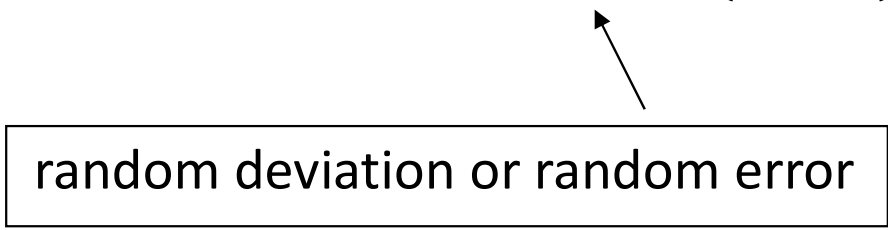
A red arrow points from the x_i term in the sample notation to the 'independent variable' box. A blue arrow points from the y_i term to the 'dependent variable' box.

independent
variable

dependent
variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

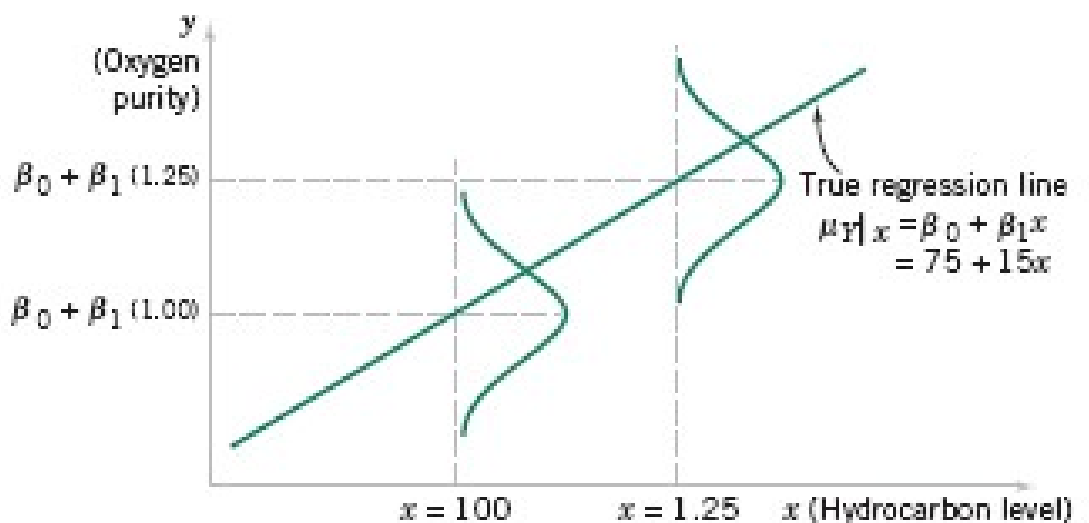
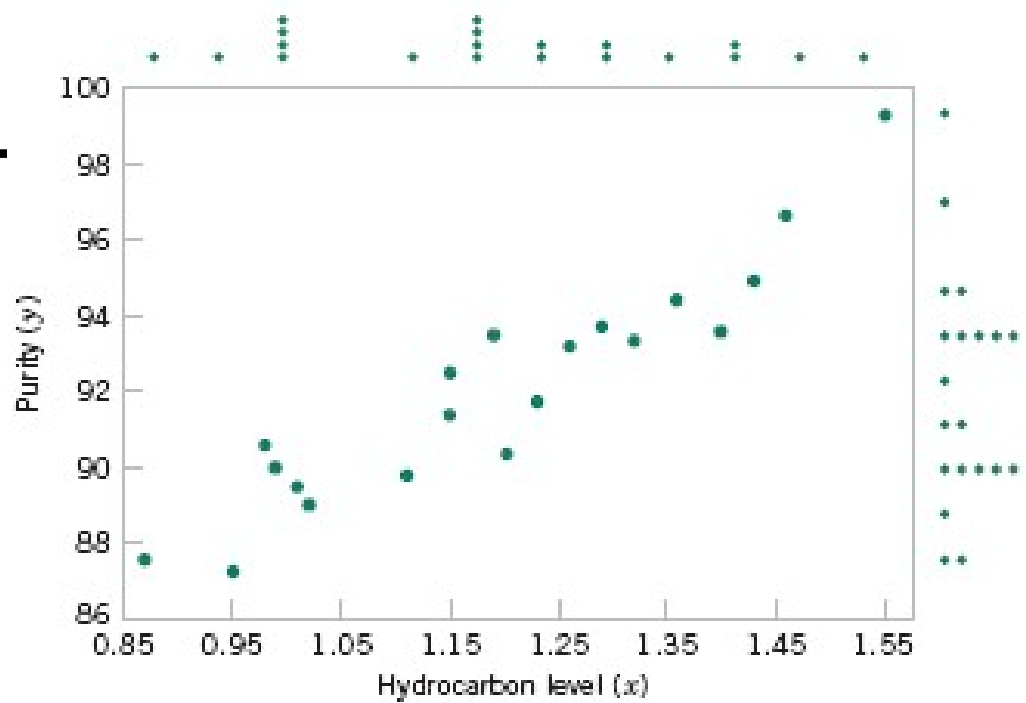


A black arrow points from the 'random deviation or random error' box to the ε term in the regression equation above.

random deviation or random error

Empirical Models

Observation Number	Hydrocarbon Level $x(\%)$	Purity $y(\%)$
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33



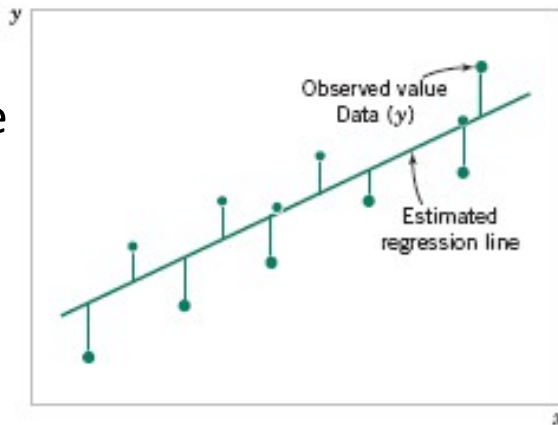
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

What are we interested in?

Building the model

- Estimating β_0 and β_1
→ the fitted regression line

b_0, b_1 : Minimize the sum of squared deviations.



Evaluating the model

- How useful is the model?
- Is the model significant?

- Estimating σ
- Distribution of $b_0, b_1, \hat{Y} = b_0 + b_1 x$

How much of the variability in Y is explained by X?

- The sample: $\{(x_i, y_i)\}_{i=1}^n$

- LR model:
$$Y = \underbrace{\beta_0 + \beta_1 X}_{\text{True regression line}} + \overset{\text{Random errors}}{\varepsilon} \quad \varepsilon \sim N(0, \sigma^2)$$

$$y_i = \underbrace{b_0 + b_1 x_i}_{\substack{\text{Estimated} \\ \text{regression line}}} + \underset{\text{residuals}}{e_i}$$

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2$$

$$SST = \sum (y_i - \bar{y})^2 = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE}$$

ANalysis Of VAriance identity

Goodness-of-fit
statistics

$$SST = \sum (y_i - \bar{y})^2 = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE}$$

$$\frac{\sum (y_i - \bar{y})^2}{SST} = \frac{\sum (\hat{y}_i - \bar{y})^2}{SST} + \frac{\sum (y_i - \hat{y}_i)^2}{SST}$$

$$1 = r^2 + \frac{SSE}{SST}$$

Coefficient of determination

- The sample: $\{(x_i, y_i)\}_{i=1}^n$
- LR model: $Y = \beta_0 + \beta_1 X + \varepsilon, \varepsilon \sim N(0, \sigma^2)$

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST} \longrightarrow$$

Proportion of observed y variation that can be explained by the simple linear regression model.
measures the degree of **linear** relationship between X and Y

Example : Computer software

Regression Analysis

The regression equation is

$$\text{Purity} = 74.3 + 14.9 \text{ HC Level}$$

Predictor	Coef	SE Coef	T	P
Constant	74.283 $\leftarrow \hat{\beta}_0$	1.593	46.62	0.000
HC Level	14.947 $\leftarrow \hat{\beta}_1$	1.317	11.35	0.000

S = 1.087

R-Sq = 87.7%

R-Sq (adj) = 87.1%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	152.13	152.13	128.86	0.000
Residual Error	18	21.25 $\leftarrow SS_E$	1.18 $\leftarrow \hat{\sigma}^2$		
Total	19	173.38			

Predicted Values for New Observations

New Obs	Fit	SE Fit	95.0% CI	95.0% PI
1	89.231	0.354	(88.486, 89.975)	(86.830, 91.632)

Values of Predictors for New Observations

New Obs	HC Level
1	1.00

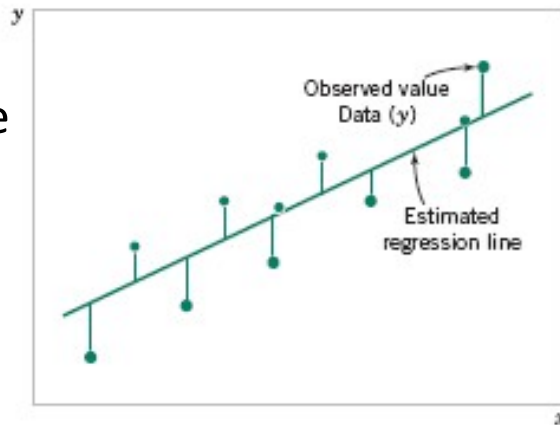
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

Is the model significant?

What are we interested in?

Building the model

- Estimating β_0 and β_1
 $\rightarrow$ the fitted regression line



b_0, b_1 : Minimize the sum of squared deviations.

$$b_0 = \bar{y} - b_1 \bar{x} \sim N \left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \right)$$

$$b_1 = \frac{S_{xy}}{S_{xx}} \sim N \left(\beta_1, \frac{\sigma^2}{S_{xx}} \right)$$

$$T = \frac{b_1 - \beta_1}{\hat{\sigma} / \sqrt{S_{xx}}} = \frac{b_1 - \beta_1}{S_{b_1}} \sim t_{n-2}$$

- Estimating σ
- Distribution of $b_0, b_1, \hat{Y} = \hat{\mu}_{Y|x_0} = b_0 + b_1 x_0$

$$\hat{\sigma}^2 = \frac{SSE}{n-2} = MSE$$

Hypothesis testing on β_1

$$H_0: \beta_1 = 0$$

$$\text{Test statistics: } T = \frac{b_1}{s_{b_1}} \sim t_{n-2}$$

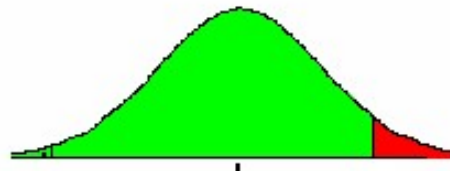
$$H_a: \beta_1 > 0$$

$$H_a: \beta_1 \neq 0$$

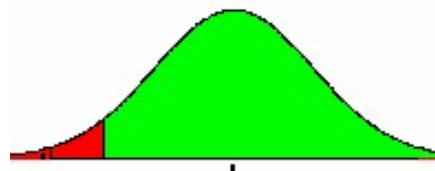
$$H_a: \beta_1 < 0$$

Accept

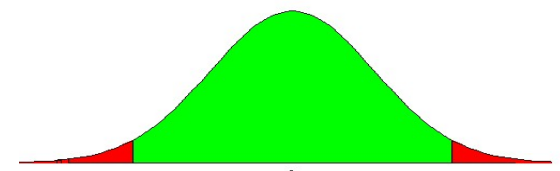
Reject



$$\text{R.R.: } t \geq t_\alpha$$



$$\text{R.R.: } t \leq -t_\alpha$$



$$\begin{aligned} \text{R.R.: } t &\leq -t_{\alpha/2} \\ \text{or } t &\geq t_{\alpha/2} \end{aligned}$$

Inferences concerning $\mu_{Y|x_0}$ (1)

$$\hat{Y} = b_0 + b_1 x_0 \quad \sim \text{normal}$$

$$E(\hat{Y}) = \beta_0 + \beta_1 x_0$$

$$V(\hat{Y}) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

$$\hat{V}(\hat{Y}) = \hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

Theorem

The variable:

$$T = \frac{b_0 + b_1 x^* - (\beta_0 + \beta_1 x^*)}{S_{b_0 + b_1 x^*}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x^*)}{S_{\hat{Y}}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x^*)}{\hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}}$$

has a t—distribution with degrees of freedom $n-2$.

Prediction interval for a future value of Y

- Error of prediction: $Y - (b_0 + b_1 x_0)$

$$E(Y - (b_0 + b_1 x_0)) = 0$$

$$V(Y - (b_0 + b_1 x_0)) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

Theorem

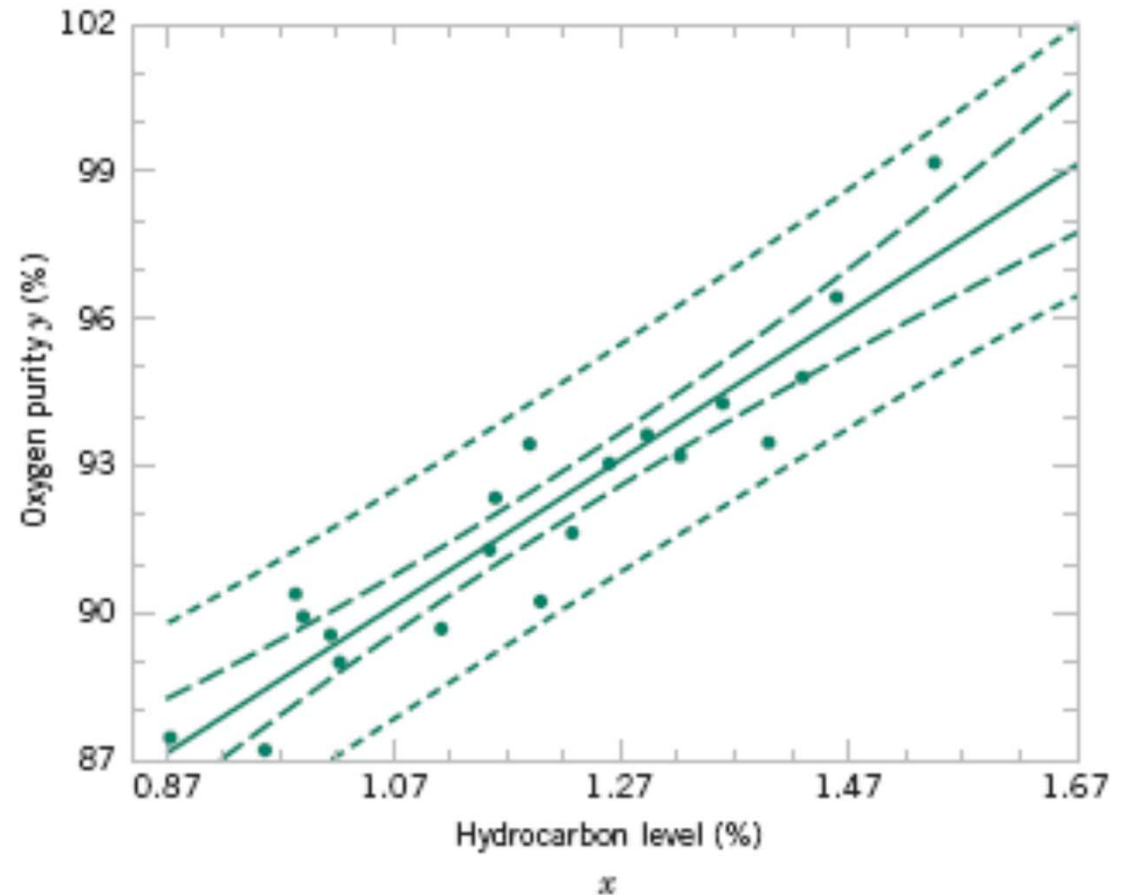
The variable:

$$T = \frac{Y - (b_0 + b_1 x_0)}{S \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} = \frac{Y - (b_0 + b_1 x_0)}{\sqrt{S^2 + S_{b_0 + b_1 x_0}^2}} = \frac{Y - (b_0 + b_1 x_0)}{\sqrt{S^2 + S_{\hat{Y}}^2}},$$

with $S = \hat{\sigma}$, has a t-distribution with degrees of freedom $n-2$.

Example

Scatter diagram of oxygen purity data with fitted regression line, 95% prediction limits (outer lines), and 95% confidence limits (inner lines) on $\mu_{Y|x_0}$.



$$\text{Mean: } T = \frac{\hat{Y} - (\beta_0 + \beta_1 x_0)}{S_{\hat{Y}}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x_0)}{\hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} \sim t_{n-2}.$$

$$\text{Prediction: } T = \frac{\hat{Y} - (\beta_0 + \beta_1 x_0)}{S_{\hat{Y}-Y}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x_0)}{\hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} \sim t_{n-2}.$$

Class Exercise

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).

Regression Analysis: Rating versus Meters

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	1669.02	1669.02	61.02	0.000
Error	30	820.50	27.35		
Total	31	2489.52			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
5.22972	67.04%	65.94%	62.11%

Coefficients

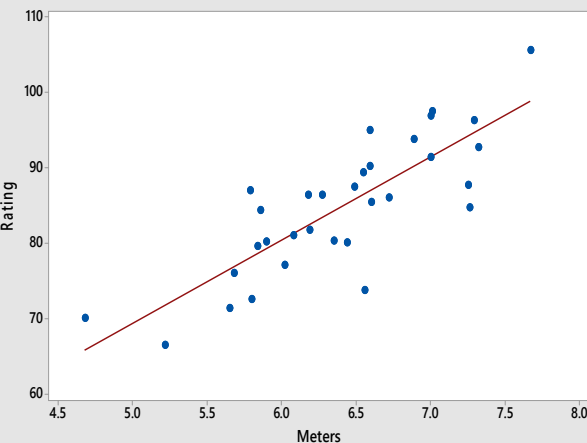
Term	Coef	SE Coef	T-Value	P-Value
Constant	14.19	9.09	1.56	0.129
Meters	11.04	1.41	7.81	0.000

Regression Equation

Rating = 14.19 + 11.04 Meters

Is this regression model significant?

Scatterplot of Rating vs Meters



Class Exercise

$$T = \frac{b_1 - \beta_1}{\hat{\sigma} / \sqrt{S_{xx}}} = \frac{b_1 - \beta_1}{S_{b_1}} \sim t_{n-2}$$

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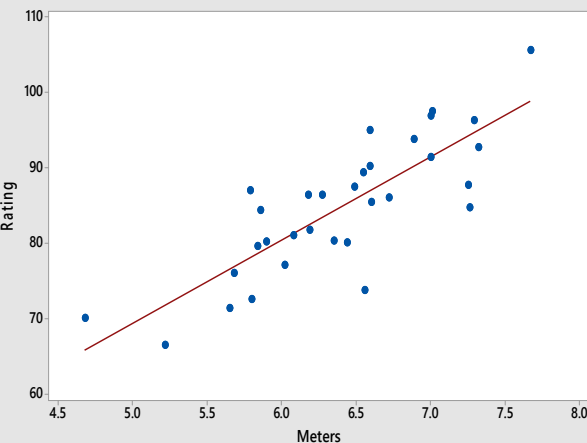
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Regression Equation

Rating = 14.19 + 11.04 Meters

Can you find a 95% confidence interval for β_1 ?

Scatterplot of Rating vs Meters



$$b_0 = \bar{y} - b_1 \bar{x}$$

Class Exercise

$$T = \frac{\hat{Y} - (\beta_0 + \beta_1 x^*)}{S_{\hat{Y}}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x^*)}{\hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} \sim t_{n-2}$$

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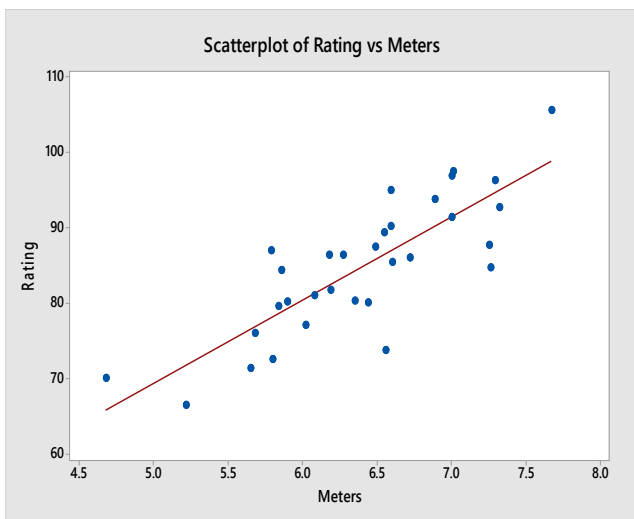
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Can you find a 95% confidence interval around Rating for Meters=7?

Class Exercise

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).

$$T = \frac{\hat{Y} - (\beta_0 + \beta_1 x^*)}{S_{\hat{Y}}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x^*)}{\hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} \sim t_{n-2}$$

Regression Analysis: Rating versus Meters

Analysis of Variance

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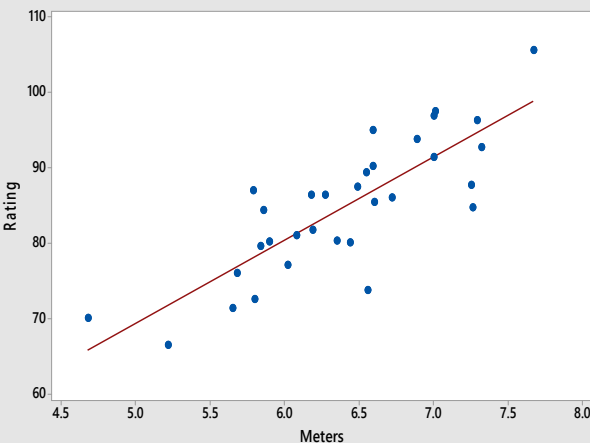
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Regression Equation

$$\text{Rating} = 14.19 + 11.04 \text{ Meters}$$

Can you find a 95% prediction interval around Rating for Meters=7?

Scatterplot of Rating vs Meters



Class Exercise

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).

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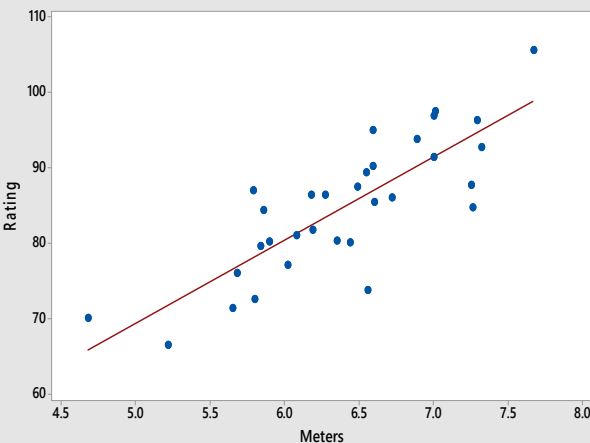
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Regression Equation

$$\text{Rating} = 14.19 + 11.04 \text{ Meters}$$

How much of the variability in Rating is explained by Meters?

Scatterplot of Rating vs Meters



Example:

A Final Exam Question

- A worldwide organization wants to estimate the literacy ratio:

LR=(number of people who can read&write)/(total number of population)

in any settlement district on earth. Their first step of analysis uses an economic indicator (EI) to generate rough predictions of the literacy ratio. The economic indicator can take continuous values in scale (0,10), where 0 is the worst possible economic situation and 10 is the best.

- Assume $\alpha=0.05$ for all questions below. For $n>60$, you can take $t_{v, 0.05}=1.67$ and $t_{v, 0.025}=2.00$.

Descriptive Statistics: LR, EI				$\sum_{i=1}^n x_i^2$ Sum of*					
				Squares	Min	Q1	Median	Q3	Max
LR	Mean 0.734	StDev 0.101	Var 0.010	43.867	0.479	0.67	0.727	0.80	0.98
EI	2.845	1.268	1.609	774.756	0.861	1.92	2.537	3.74	5.59

Example:

A Final Exam Question

Regression Analysis: LR versus EI

The regression equation is

$$\text{LR} = 0.622 + 0.0392 \text{ EI}$$

Predictor	Coef	SE Coef	T	P
Constant	0.62206	0.02445	25.44	0.000
EI	0.039221	0.007856	4.99	0.006

$S = 0.0885719$ $R\text{-Sq} = 24.2\%$ $R\text{-Sq}(\text{adj}) = 23.2\%$

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	0.19551	0.19551	24.92	0.006
Residual Error	78	0.61191	0.00784		
Total	79	0.80742			

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

Questions

- a) How much of the variability in literacy ratio is explained by the economic indicator? Would you like to improve on this? Why? If you want to improve, explain how.

Example:

A Final Exam Question

Regression Analysis: LR versus EI

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$$\text{LR} = 0.622 + 0.0392 \text{ EI}$$

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Questions

- a) How much of the variability in literacy ratio is explained by the economic indicator? Would you like to improve on this? Why? If you want to improve, explain how.
- b) You are interested in predicting the literacy ratio with an economic indicator of 4.5. If you can, find a prediction value, construct a 95% prediction interval around your prediction, and explain the meaning of this PI. If you cannot, explain why not. What changes if you are interested in LR when $EI=7.5$? Can you find a prediction interval? Why or why not? Explain.

Example:

A Final Exam Question

- A worldwide organization wants to estimate the literacy ratio:

LR=(number of people who can read&write)/(total number of population)

in any settlement district on earth. Their first step of analysis uses an economic indicator (EI) to generate rough predictions of the literacy ratio. The economic indicator can take continuous values in scale (0,10), where 0 is the worst possible economic situation and 10 is the best.

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$$\sum_{i=1}^n x_i^2$$

Sum of*

Descriptive Statistics: LR, EI

	Mean	StDev	Var	Squares	Min	Q1	Median	Q3	Max
LR	0.734	0.101	0.010	43.867	0.479	0.67	0.727	0.80	0.98
EI	2.845	1.268	1.609	774.756	0.861	1.92	2.537	3.74	5.59

Example:

$$\text{Prediction: } T = \frac{\hat{Y} - (\beta_0 + \beta_1 x_0)}{S_{\hat{Y}-Y}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x_0)}{\hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} \sim t_{n-2}.$$

A Final Exam Question

Regression Analysis: LR versus EI

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EI	0.039221	0.007856	4.99	0.006

S = 0.0885719 R-Sq = 24.2% R-Sq(adj) = 23.2%

Analysis of Variance

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Questions

- a) How much of the variability in literacy ratio is explained by the economic indicator? Would you like to improve on this? Why? If you want to improve, explain how.
- b) You are interested in predicting the literacy ratio with an economic indicator of 4.5. If you can, find a prediction value, construct a 95% prediction interval around your prediction, and explain the meaning of this PI. If you cannot, explain why not. What changes if you are interested in LR when EI=7.5? Can you find a prediction interval? Why or why not? Explain.
- c) State the appropriate hypotheses to test whether literacy ratio increases with the economic indicator or not, i.e. write down the corresponding H_0 and H_a . Find the p-value of the test and discuss its meaning. What is your conclusion for the test? Explain.

$$H_0: \beta_1 = 0$$

$$H_a: \beta_1 > 0$$

- **Test statistics?**
- **P-value?**

Example:

A Final Exam Question

Regression Analysis: LR versus EI

The regression equation is

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Total	79	0.80742			

Linear regression model

- The sample: $\{(x_i, y_i)\}_{i=1}^n$
- LR model:
$$Y = \underbrace{\beta_0 + \beta_1 X}_{\text{True regression line}} + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$
- Assumptions:
 - Random errors are normal.
 - Random errors are independent of x and y values as well as the order i.
 - Variance of errors is constant.
 - There exists a linear relationship between x and y.

Adequacy of the Regression Model

- Observation: Y_i
- Estimate: $\hat{Y}_i = b_0 + b_1 x_i$
- Residual: $e_i = Y_i - \hat{Y}_i$
- Variance of residual:

$$V(Y_i - \hat{Y}_i) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}} \right)$$

↓

Replaced by MSE

Adequacy of the Regression Model

- Estimated variance of residual:

$$\hat{V}(Y_i - \hat{Y}_i) = \hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(x_i - \bar{x}_i)^2}{S_{xx}} \right)$$

- But standardized residual can be computed differently in different books, softwares etc.:

$$e_i^* = \frac{y_i - \hat{y}_i}{\hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_i - \bar{x}_i)^2}{S_{xx}}}} \quad \text{or} \quad e_i^* = \frac{y_i - \hat{y}_i}{\hat{\sigma}} \quad i = 1, \dots, n$$

Diagnostic Plots

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

Assumptions on ε_i :

- | | | |
|---|-----------------------|--|
| 1) e_i^* (or e_i) versus x_i | $\longleftrightarrow$ | 1) ε_i 's (so $E(\varepsilon_i)$ and $V(\varepsilon_i)$) do not depend on x_i . |
| 2) e_i^* (or e_i) versus y_i | $\longleftrightarrow$ | 2) ε_i 's (so $E(\varepsilon_i)$ and $V(\varepsilon_i)$) do not depend on y_i . |
| 3) normal probability plot of e_i^* (or e_i) | $\longleftrightarrow$ | 3) ε_i 's are normal. |
| 4) e_i^* (or e_i) versus i | | 4) ε_i 's are independent over time, i.e., do not depend on i . |

Diagnostic Plots

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

5) $\hat{y}_i$ versus y_i



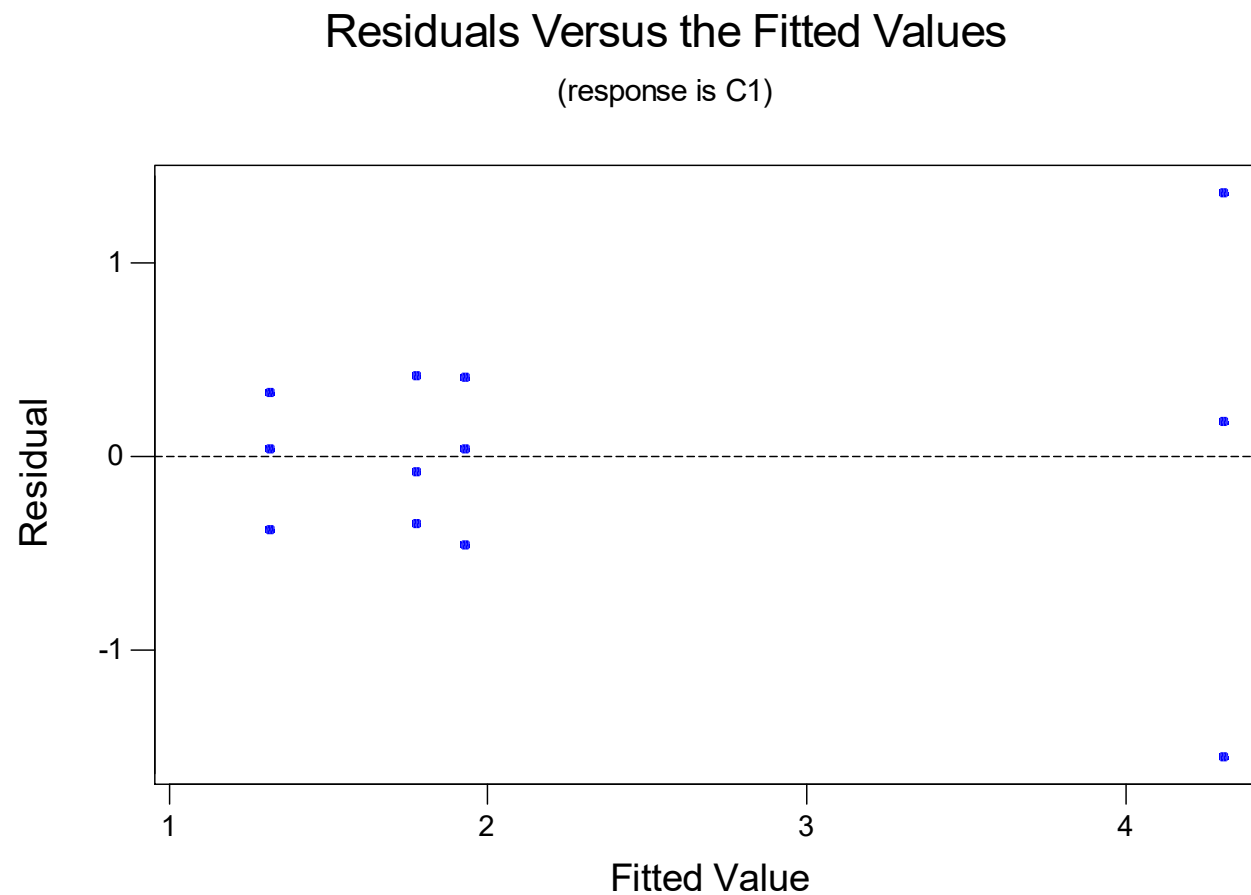
5) close to a 45° line??

6) y_i versus x_i



6) an observation with large influence

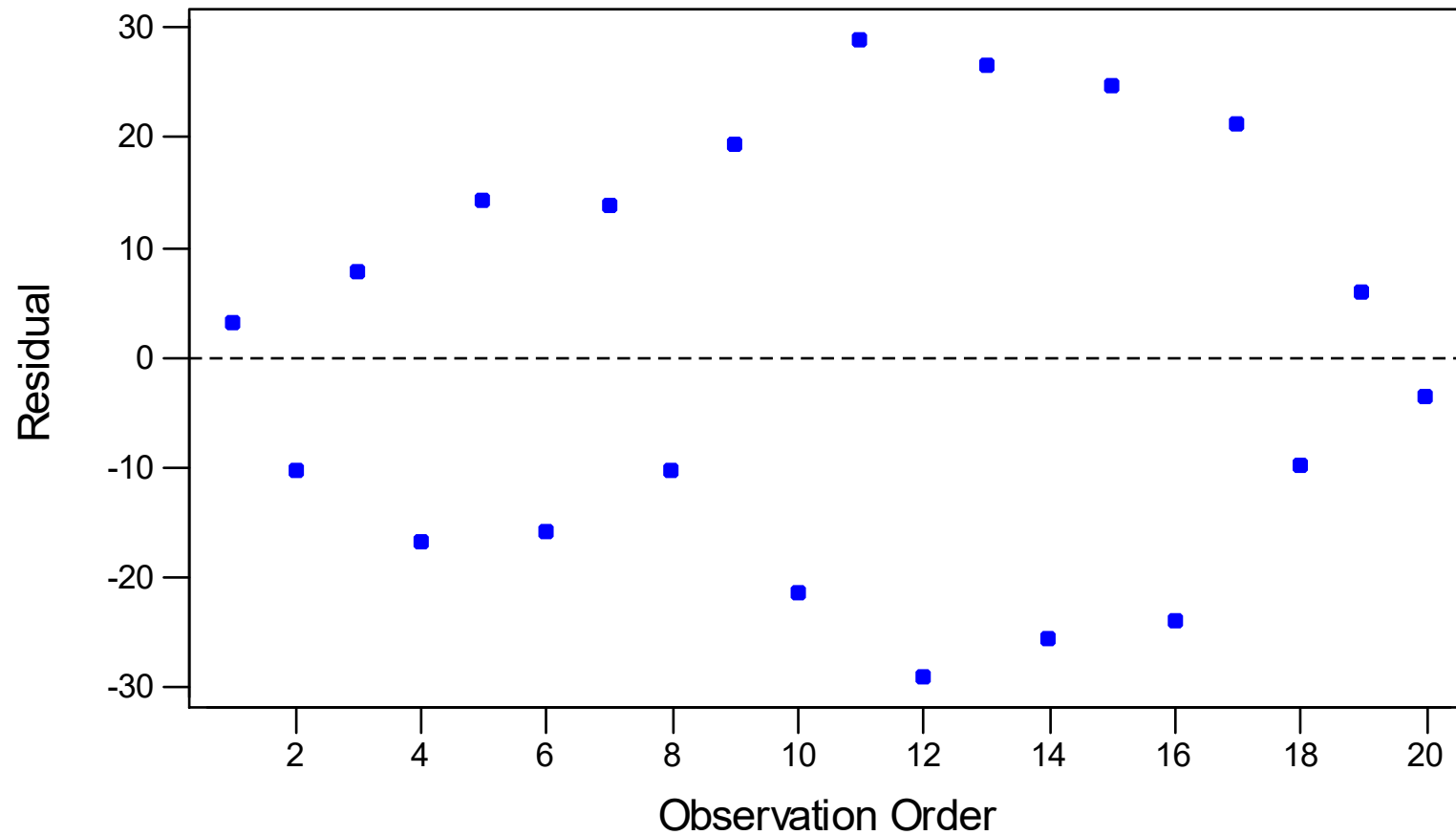
A change in the variance of errors with respect to the fitted values



Autocorrelation

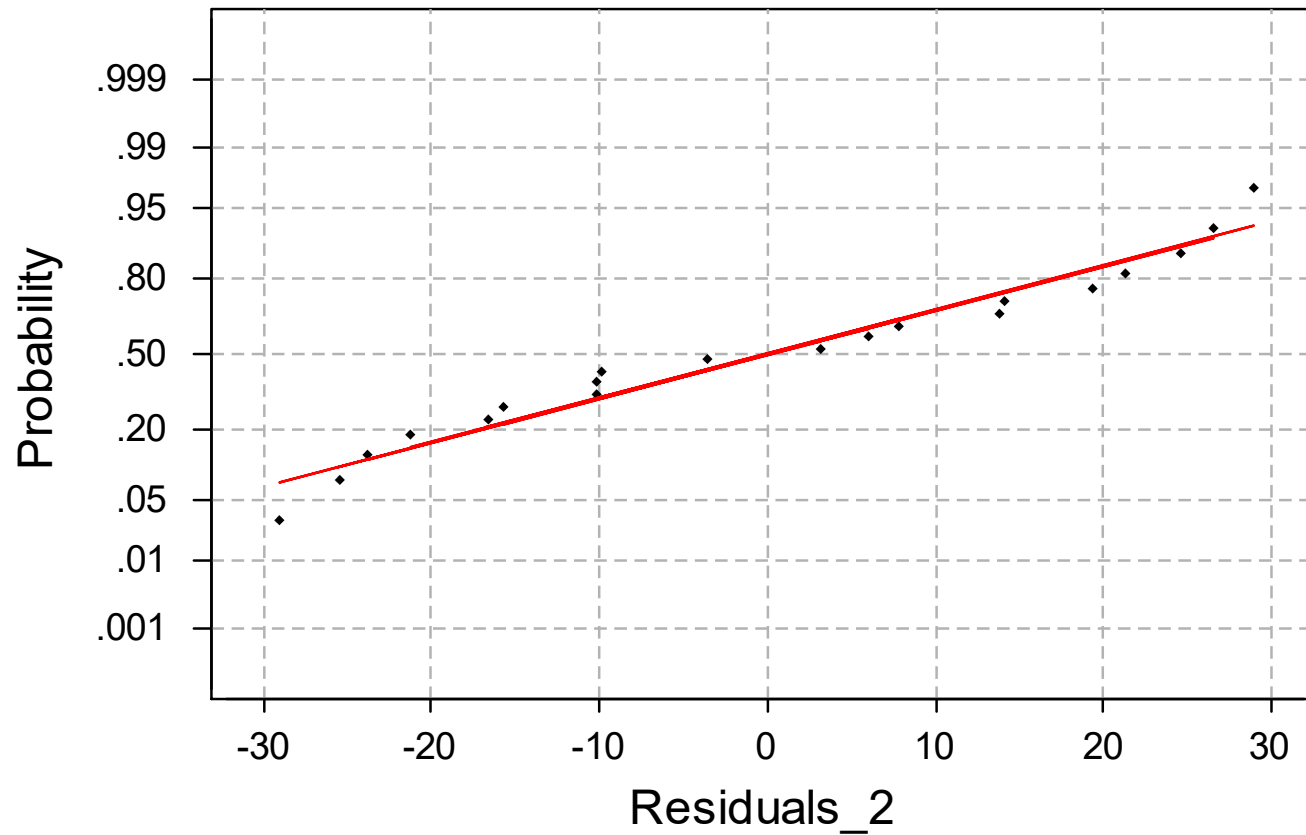
Residuals Versus the Order of the Data

(response is Actual Y)



Autocorrelation

Normal Probability Plot



Average: 0.0000500
StDev: 18.9933
N: 20

W-test for Normality
R: 0.9778
P-Value (approx): > 0.1000

Possible problems which can be detected by diagnostic plots

- A nonlinear probabilistic relationship between x and y
- Variance of ε_i is not constant, but depends on x (or y).
- Influential observations, outliers, etc. which affect the regression line a great deal
- The error term ε_i does not have a normal distribution.
- ε_i 's exhibit dependence over time.
- Lack-of-fit in regression line which can be tested statistically only when there are more than one observation for each x_i .

Transformation to a straightline

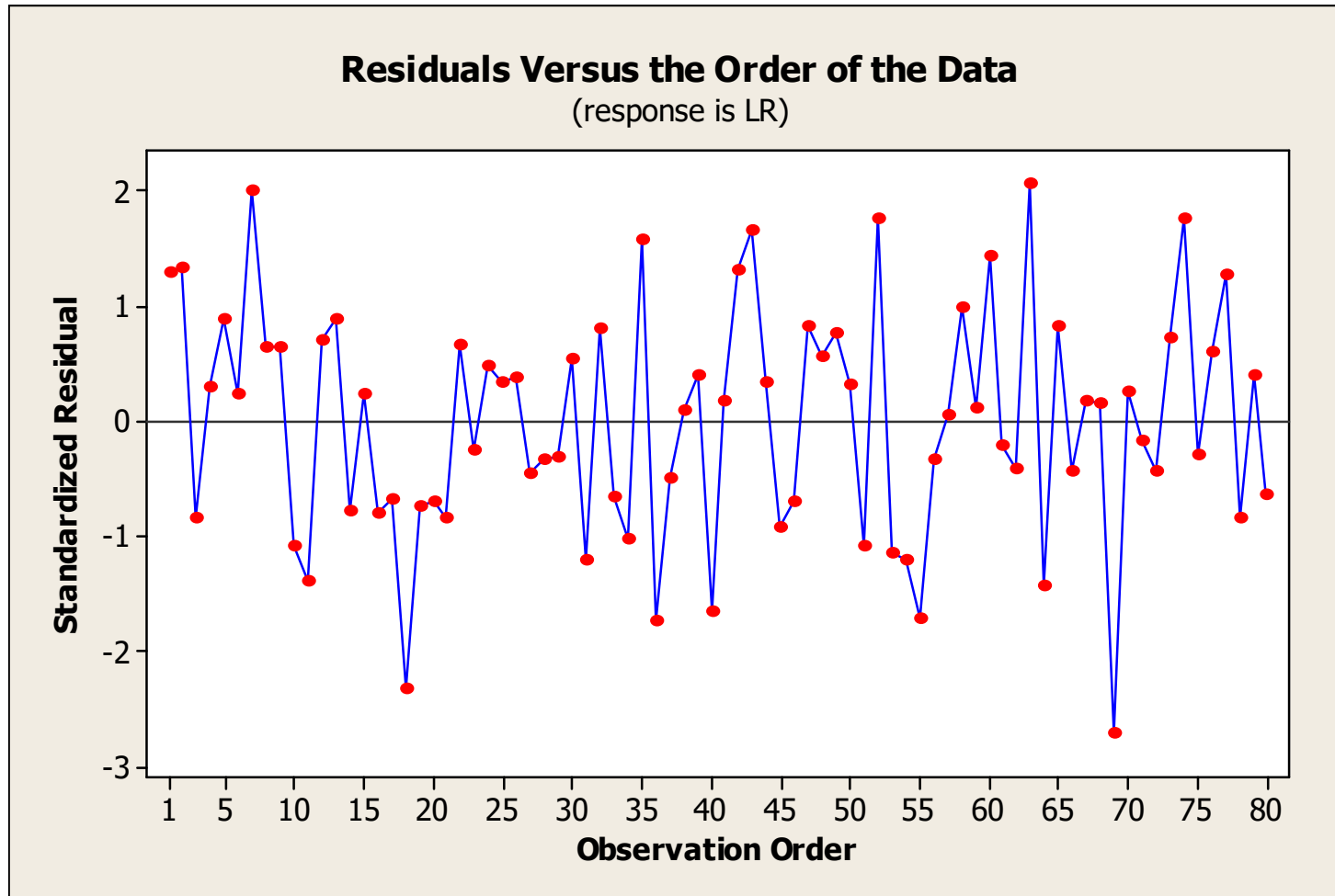
Example: $Y = \beta_0 e^{\beta_1 x} \varepsilon$

Linear regression model

- LR equation: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ $\varepsilon_i \sim N(0, \sigma^2)$
- Assumptions:
 - Random errors are normal.
 - Random errors are independent of x and y values as well as the order i.
 - Variance of errors is constant.
 - There exists a linear relationship between x and y.
- Validate the assumptions!

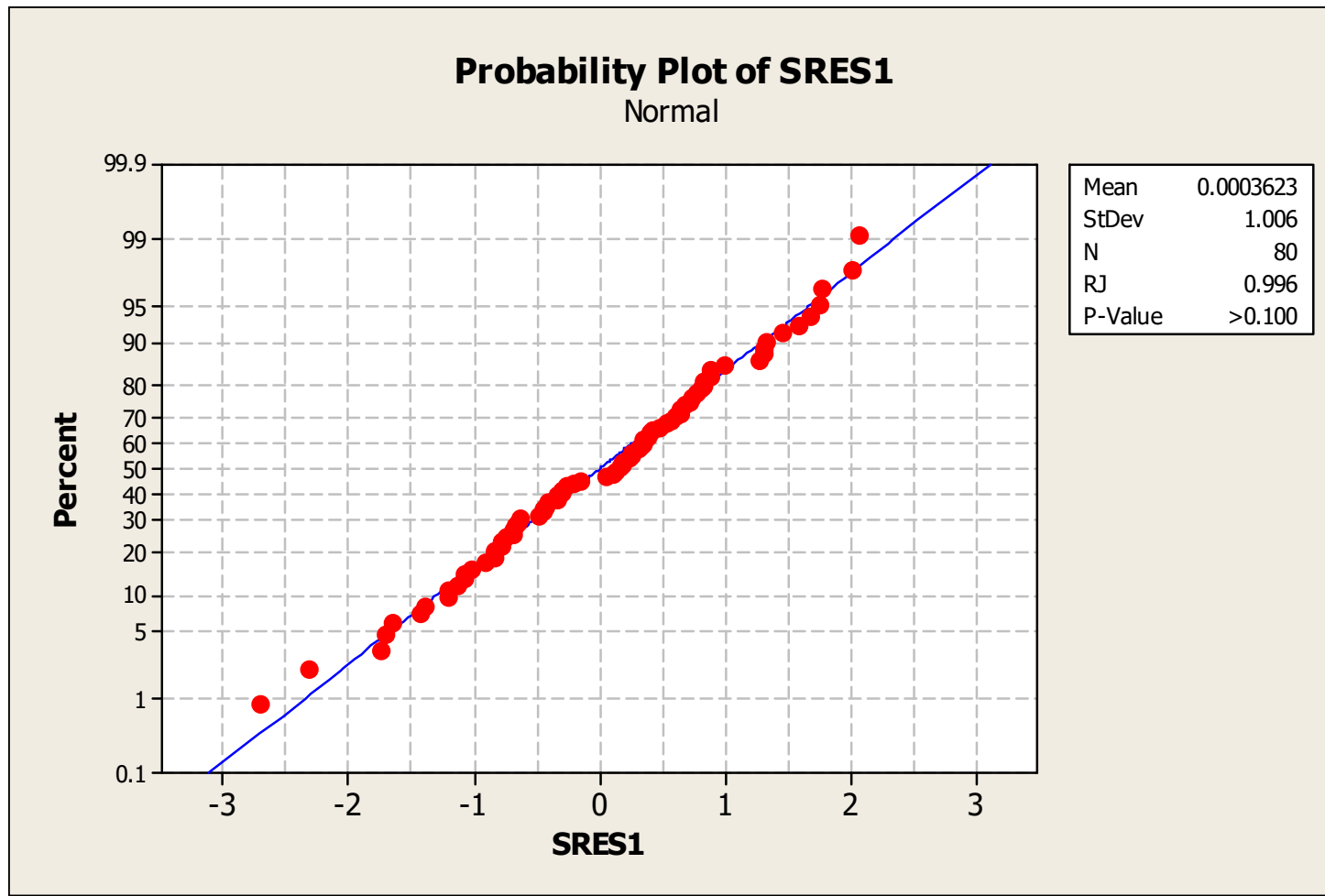
Example:

A Final Exam Question



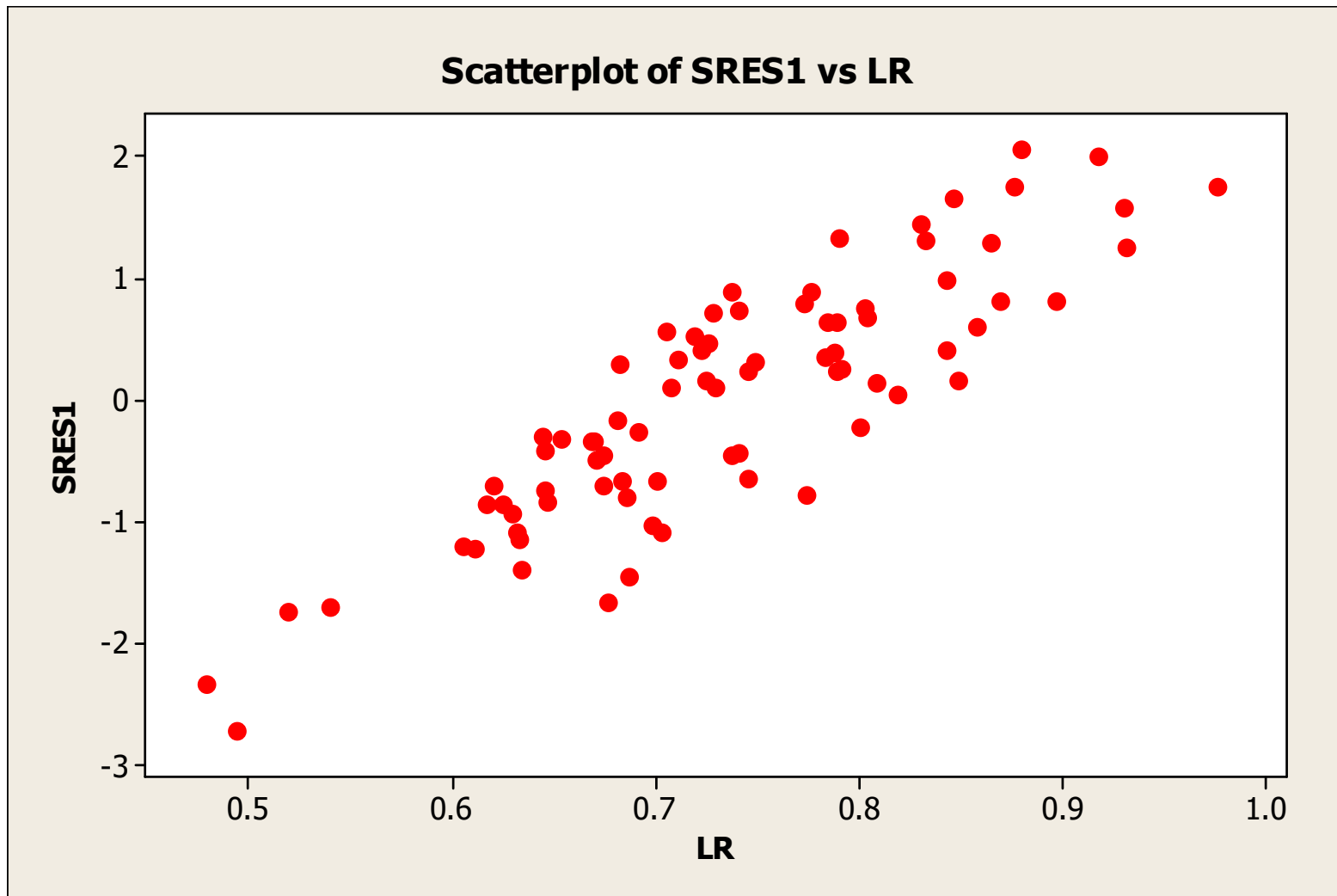
Example:

A Final Exam Question



Example:

A Final Exam Question



More examples on

- Validating the assumptions of linear regression model

Example

- 4 data sets:

Data set	1-3	1	2	3	4	4
Variable	x	y	y	y	x	y
	10.00	8.04	9.14	7.46	8.00	6.58
	8.00	6.95	8.14	6.77	8.00	5.76
	13.00	7.58	8.74	12.74	8.00	7.71
	9.00	8.81	8.77	7.11	8.00	8.84
	11.00	8.33	9.26	7.81	8.00	8.47
	14.00	9.96	8.10	8.84	8.00	7.04
	6.00	7.24	6.13	6.08	8.00	5.25
	4.00	4.26	3.10	5.39	19.00	12.50
	12.00	10.84	9.13	8.15	8.00	5.56
	7.00	4.82	7.26	6.42	8.00	7.91
	5.00	5.68	4.74	5.73	8.00	6.89

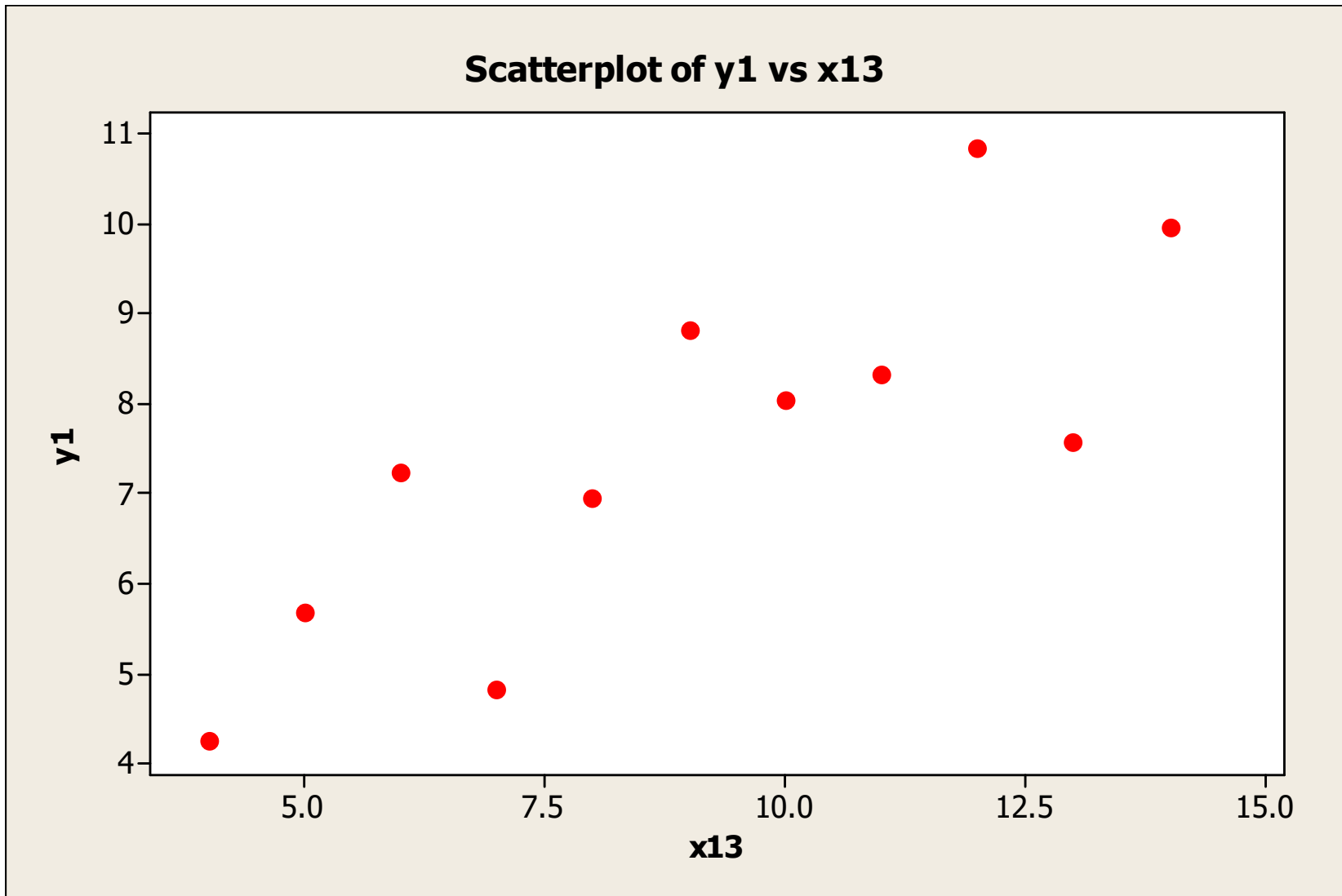
- The summary statistics for all sets are identical, hence the regression line, F and t statistics will all the same, but...

Example: Set 1

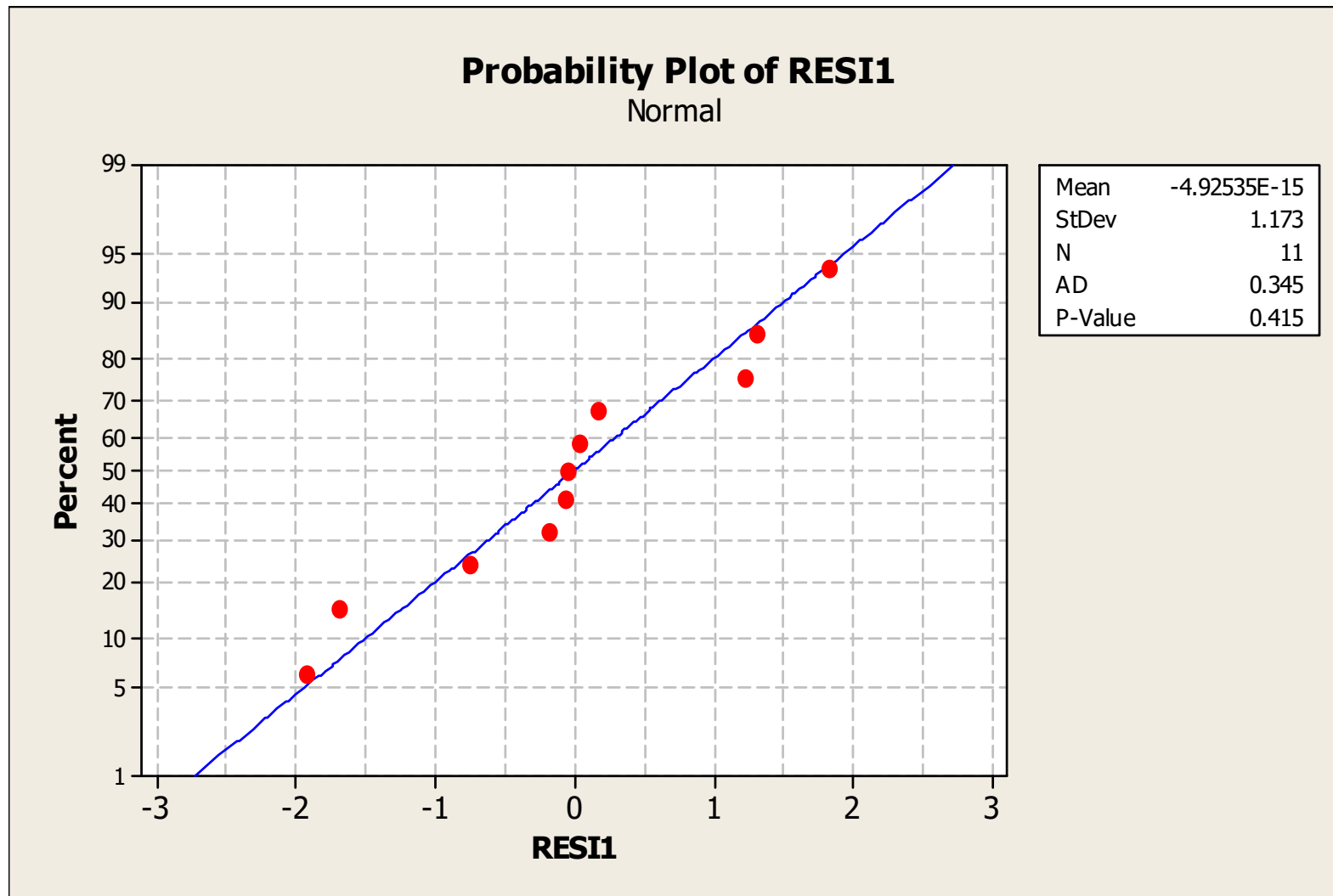
The regression equation is: $y_1 = 3.00 + 0.500 x_{13}$

Predictor	Coef	SE Coef	T	P	
Constant	3.00	1.125	2.67	0.026	
x	0.50	0.1179	4.24	0.002	
S = 1.2366		R-Sq = 66.7%		R-sq-adj 62.9%	
ANOVA					
Source	df	SS	MS	F	P
Regression	1	27.510 SSR	27.510 MSR	17.99	0.002
Error	9	13.763 SSE	1.529 MSE		
Total	10	41.273 SST			

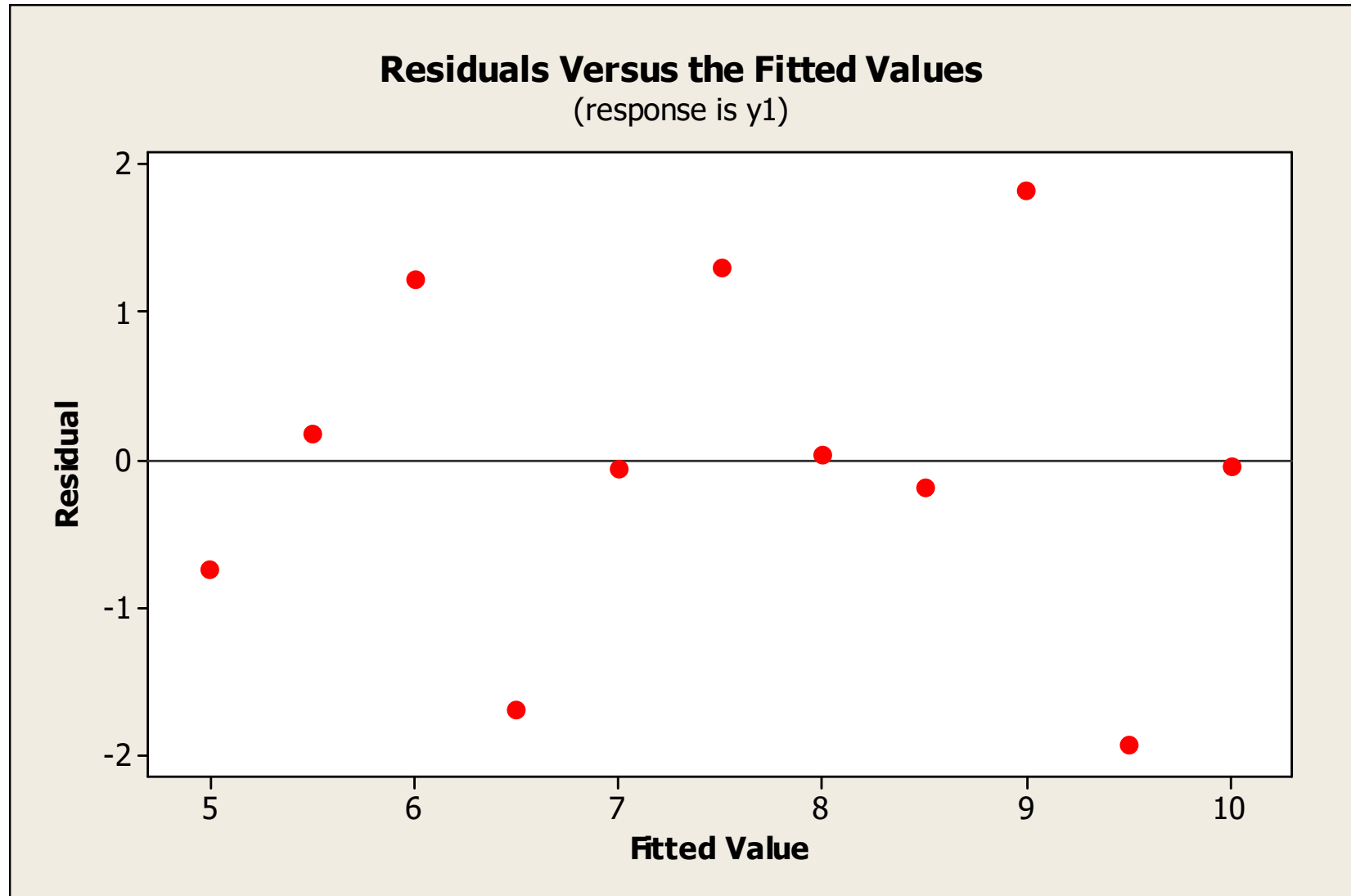
Example: Set 1



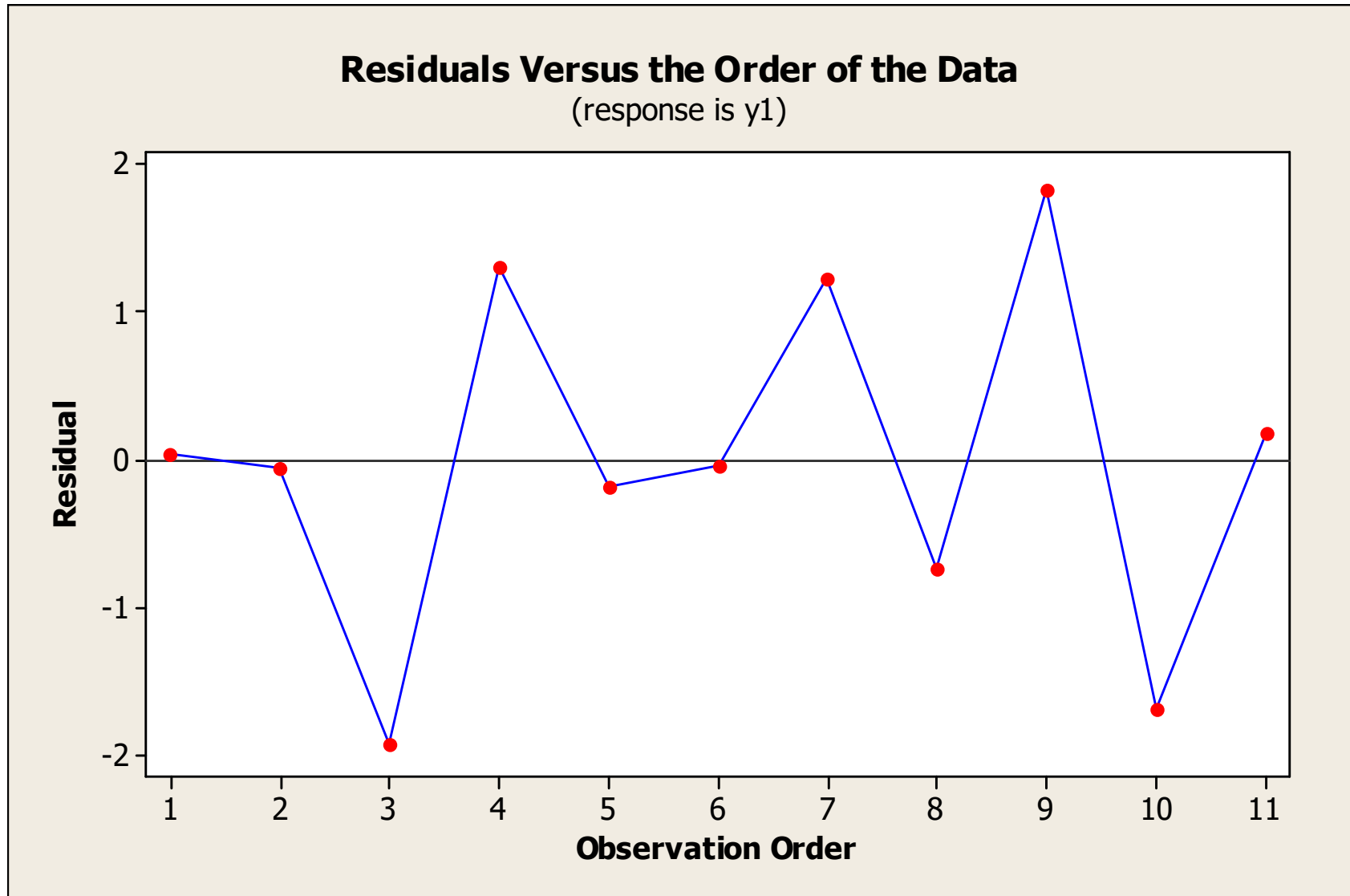
Example: Set 1



Example: Set 1



Example: Set 1

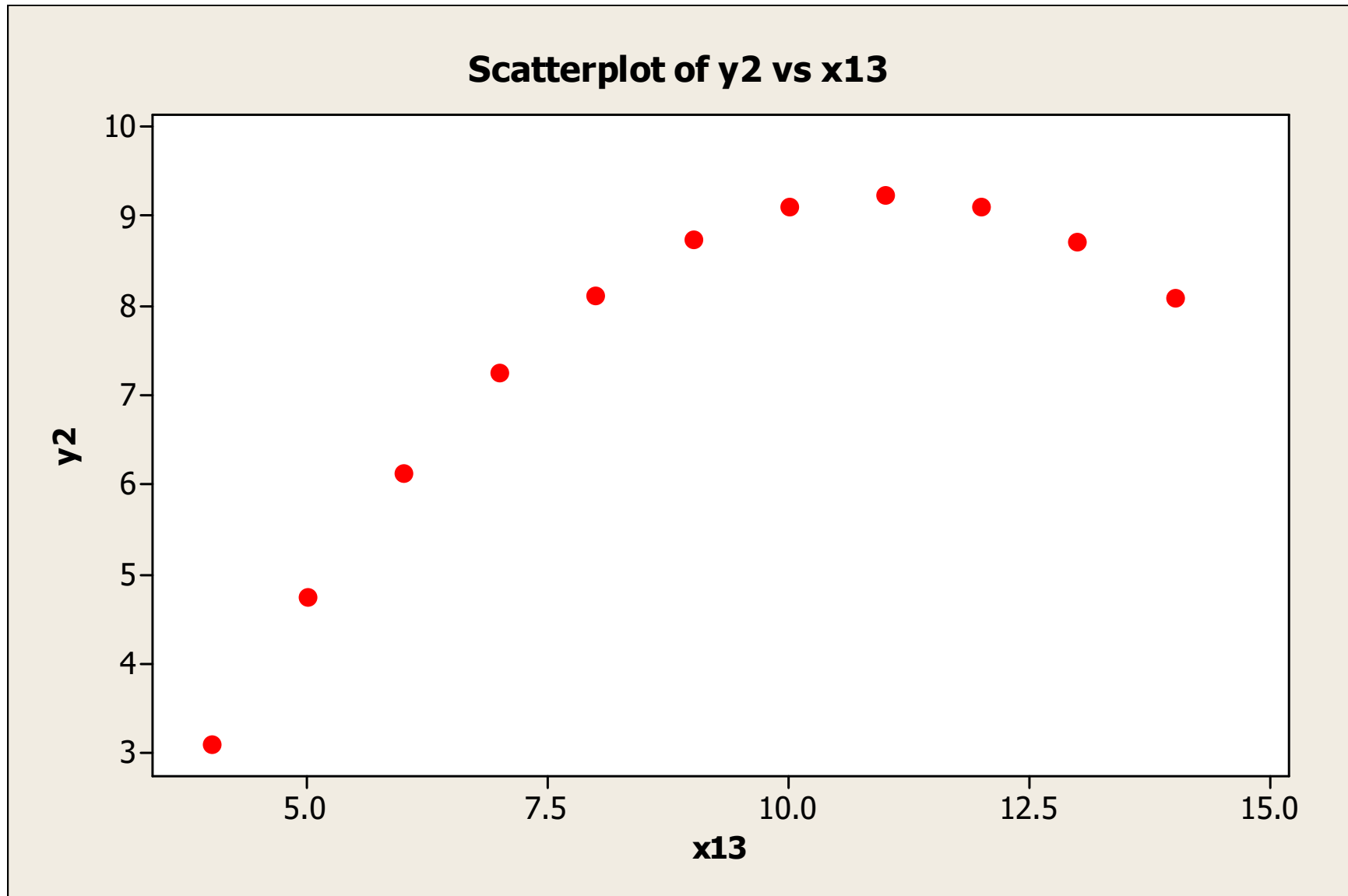


Example: Set 2

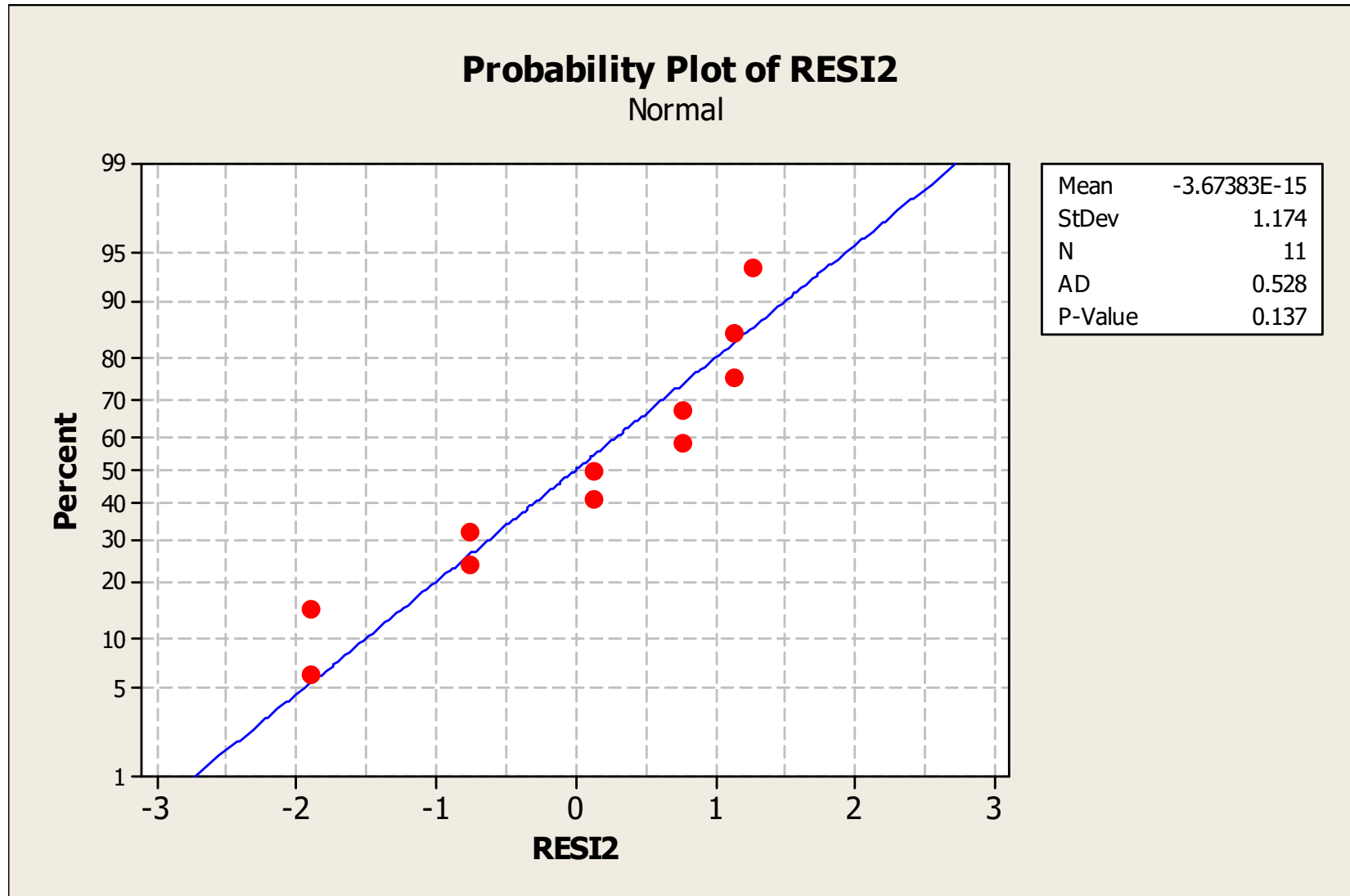
The regression equation is: $y_2 = 3.00 + 0.500 x_1$

Predictor	Coef	SE Coef	T	P	
Constant	3.00	1.125	2.67	0.026	
x	0.50	0.1179	4.24	0.002	
S = 1.2366		R-Sq = 66.7%		R-sq-adj 62.9%	
ANOVA					
Source	df	SS	MS	F	P
Regression	1	27.510	27.510	17.99	0.002
Error	9	13.763	1.529		
Total	10	41.273			

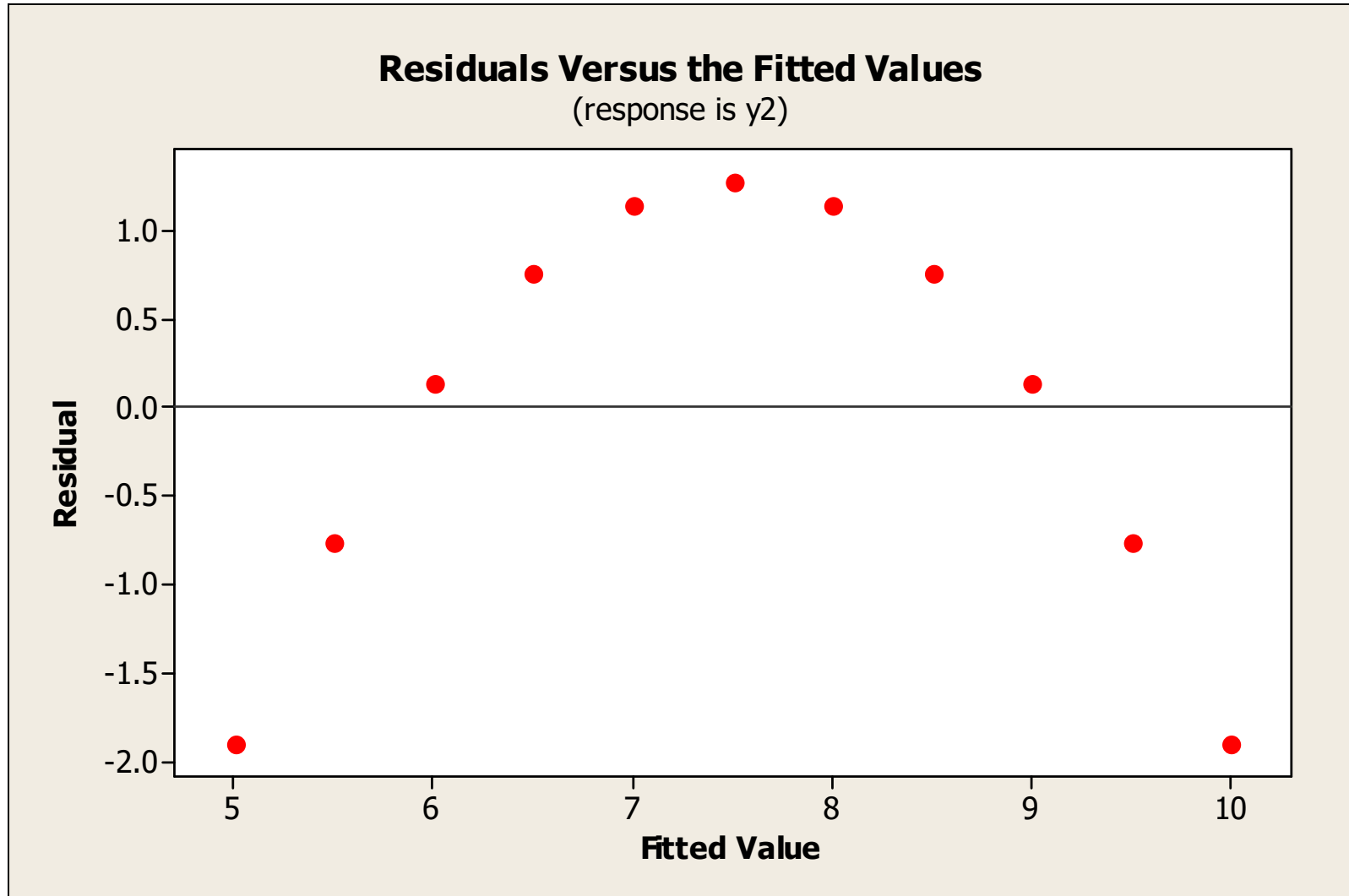
Example: Set 2



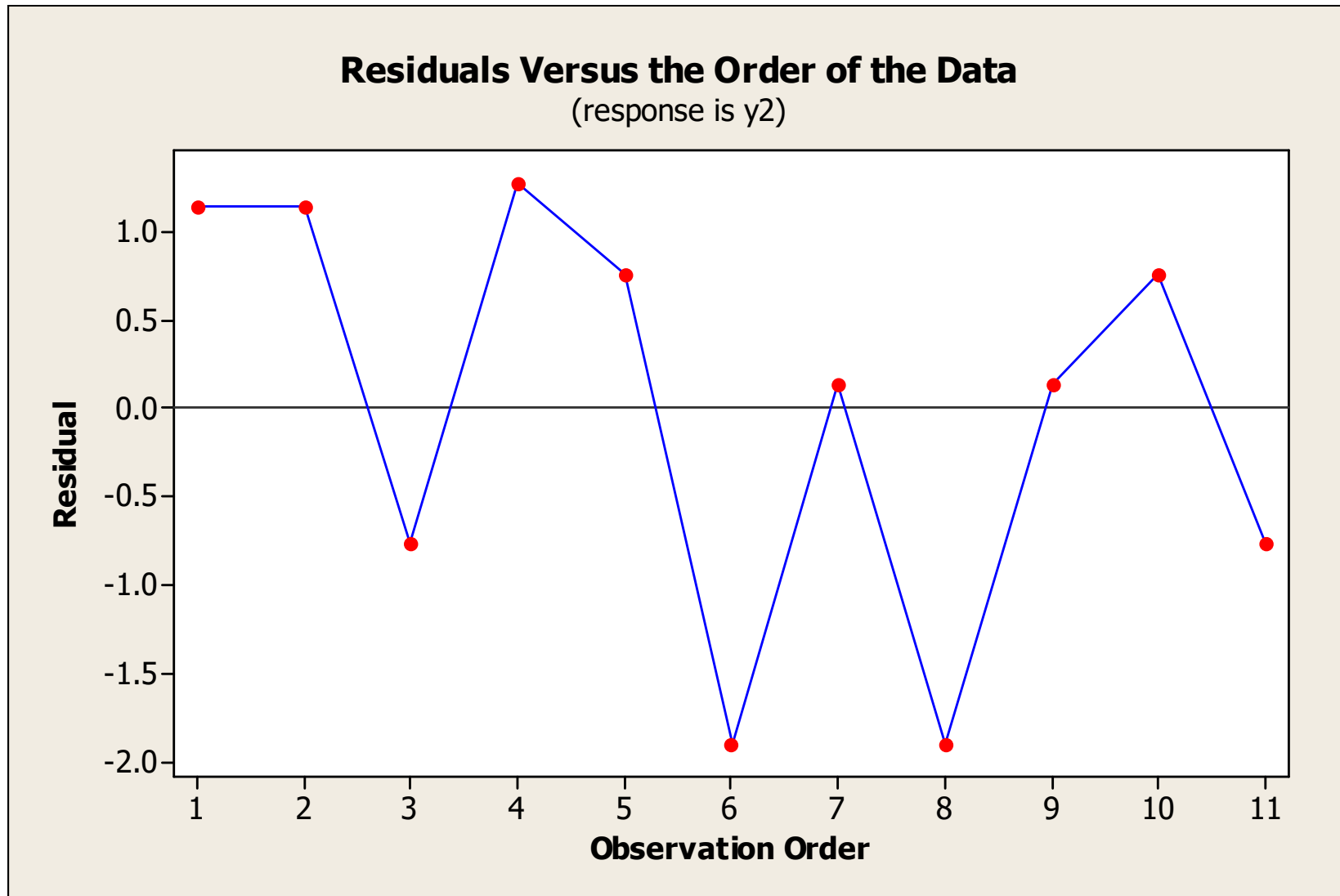
Example: Set 2



Example: Set 2



Example: Set 2



Example: Set 3

with an influential observation

$$y = 3.00 + 0.500 x$$

Predictor	Coef	SE Coef	T	P
Constant	3.002	1.124	2.67	0.026
x	0.4997	0.1179	4.24	0.002

S = 1.236 R-Sq = 66.6% R-Sq(adj) = 62.9%

Analysis of Variance

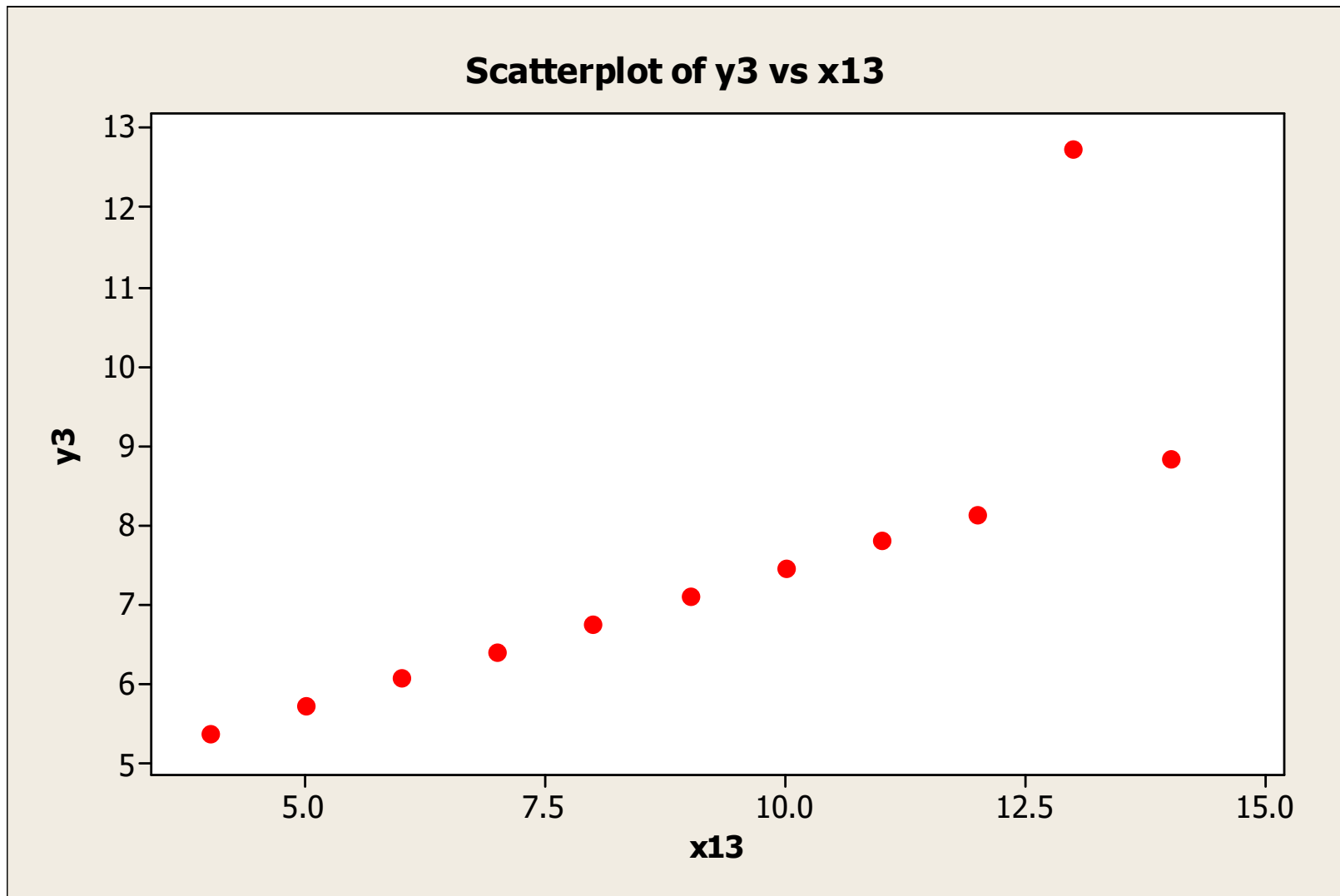
Source	DF	SS	MS	F	P
Regression	1	27.470	27.470	17.97	0.002
Residual Error	9	13.756	1.528		
Total	10	41.226			

Unusual Observations

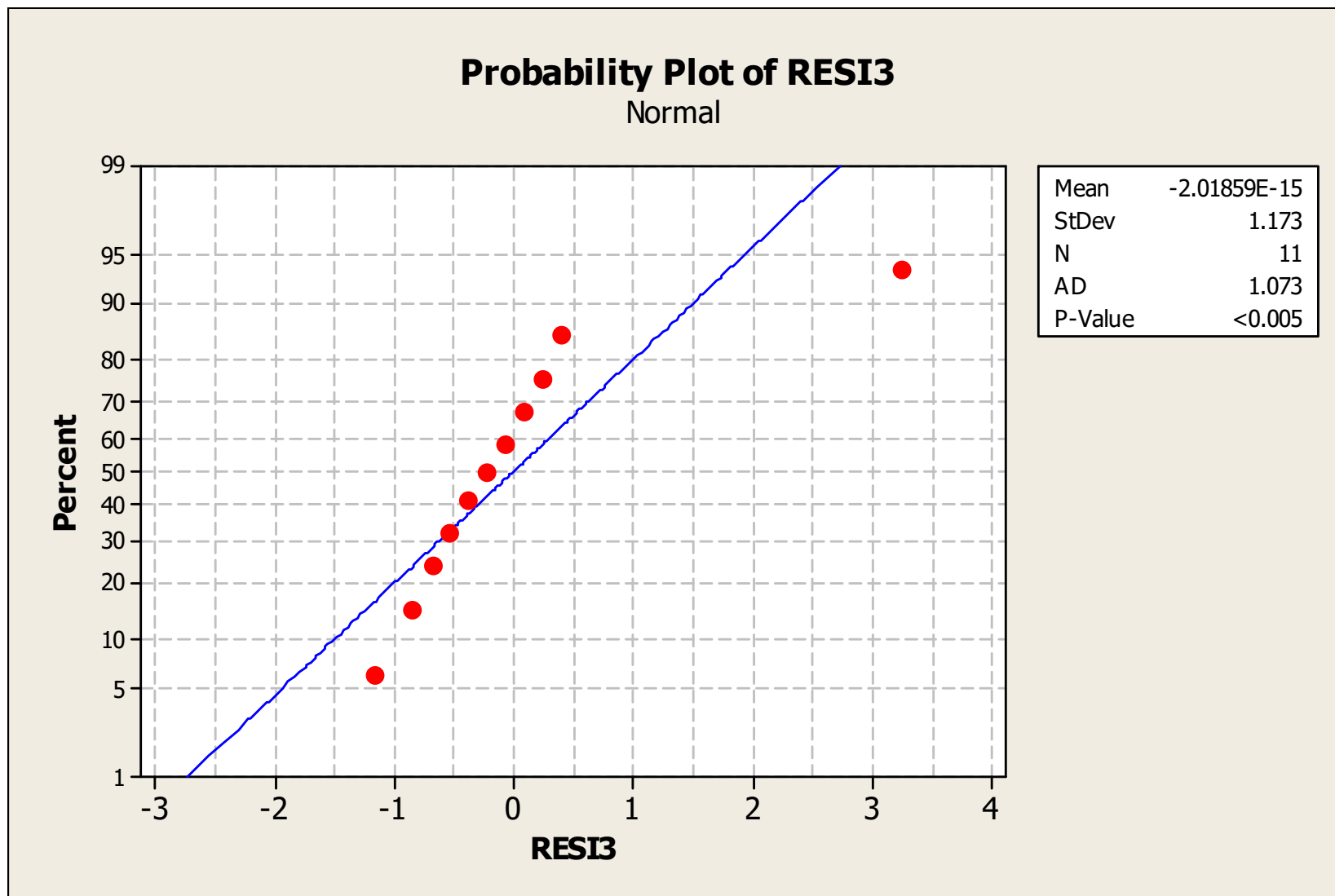
Obs	x	y	Fit	SE Fit	Residual	St.R.
3	13.0	12.740	9.499	0.601	3.241	3.00R

R denotes an observation with a large standardized residual

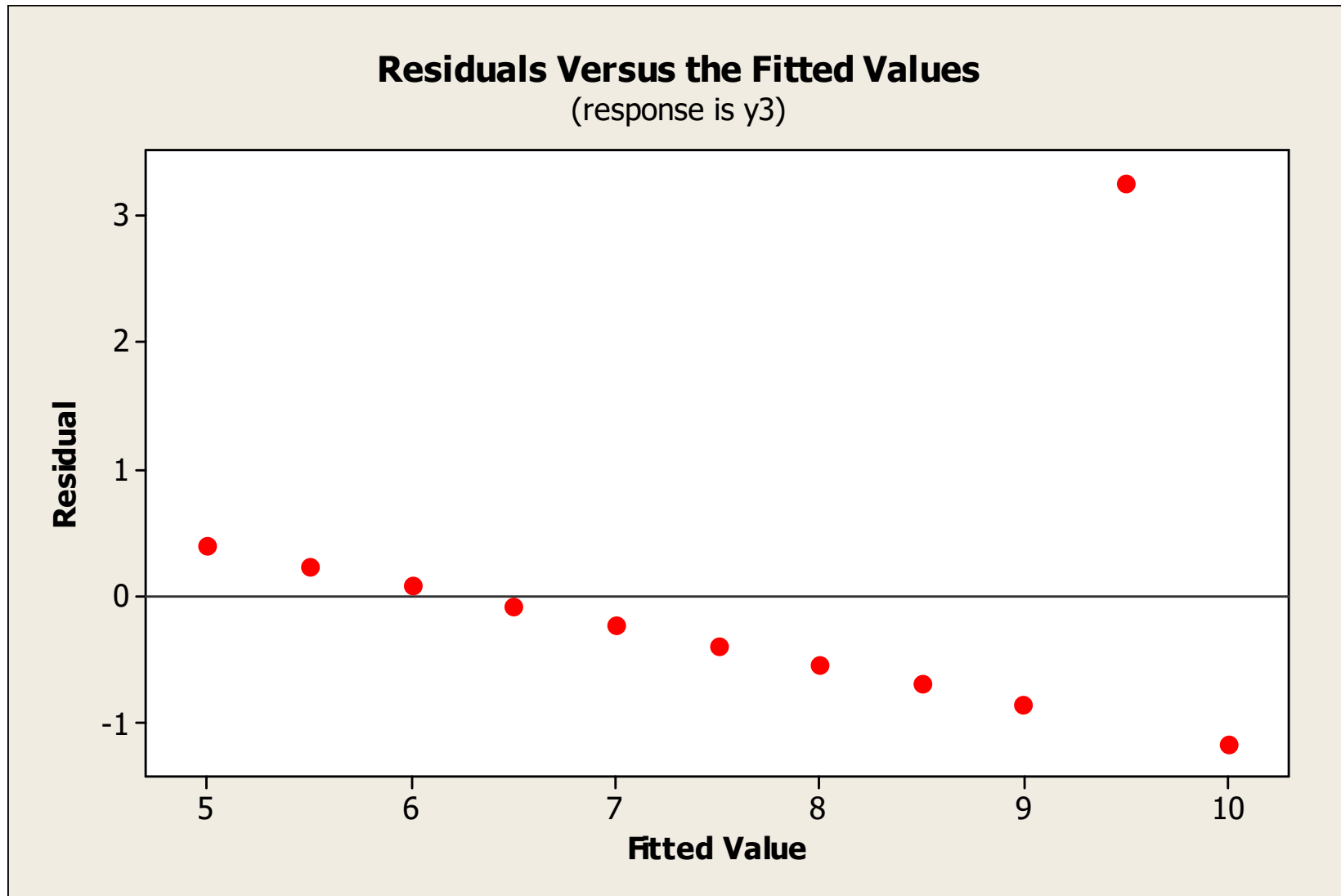
Example: Set 3



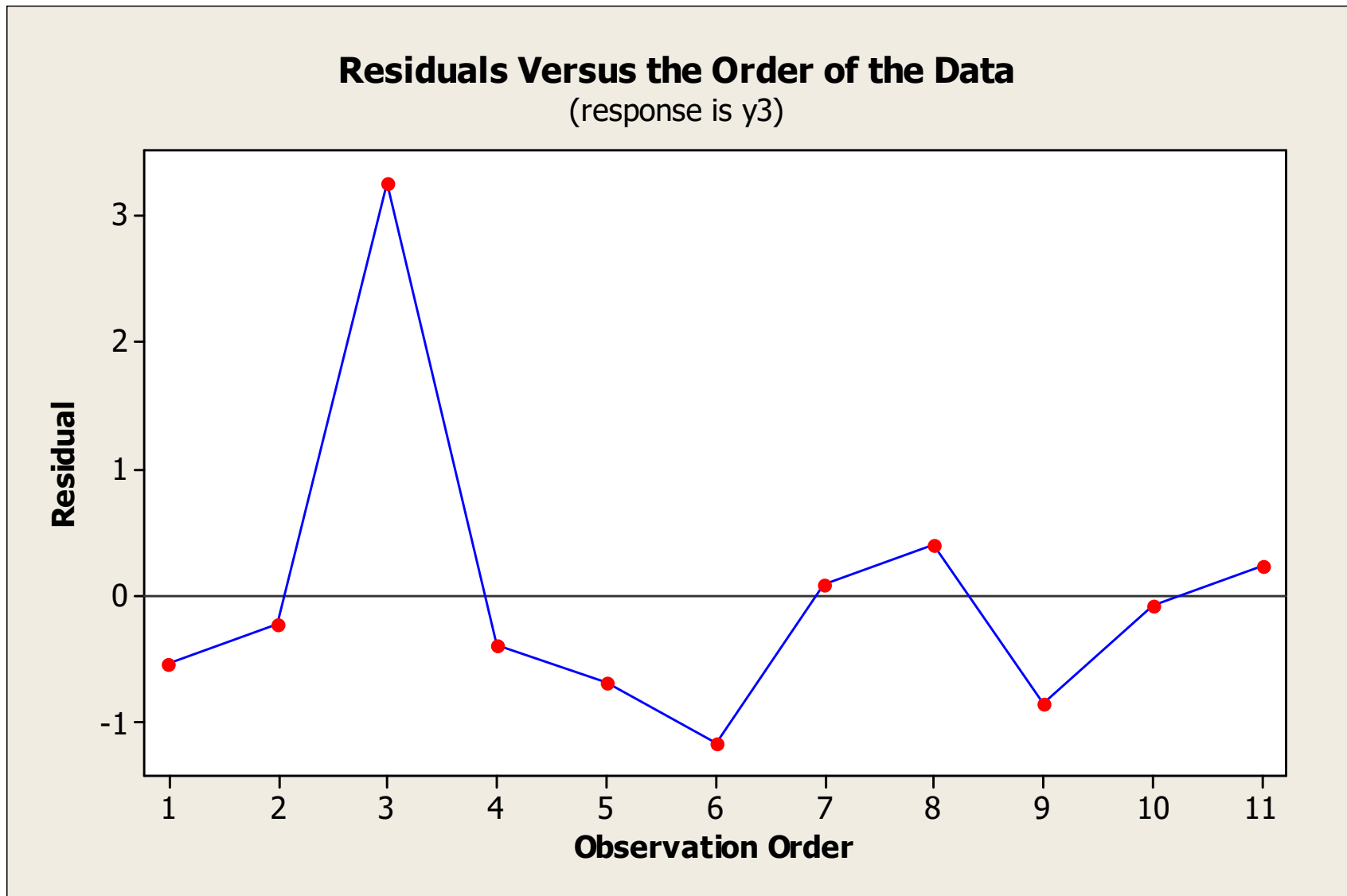
Example: Set 3



Example (set 3) with an influential observation



Example: Set 3 with an influential observation



Example: Set 3

without the influential observation

$$y' = 4.01 + 0.345 x'$$

Predictor	Coef	SE Coef	T	P
Constant	4.00565	0.00292	1369.81	0.000
x'	0.345390	0.000321	1077.35	0.000

S = 0.003082 R-Sq = 100.0% R-Sq(adj) = 100.0%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	11.023	11.023	1.161E+06	0.000
Residual Error	8	0.000	0.000		
Total	9	11.023			

Example (set 4) with an influential observation

$$y = 3.00 + 0.500 x$$

Predictor	Coef	SE Coef	T	P
Constant	3.002	1.124	2.67	0.026
x	0.4999	0.1178	4.24	0.002

S = 1.236 R-Sq = 66.7% R-Sq(adj) = 63.0%

Analysis of Variance

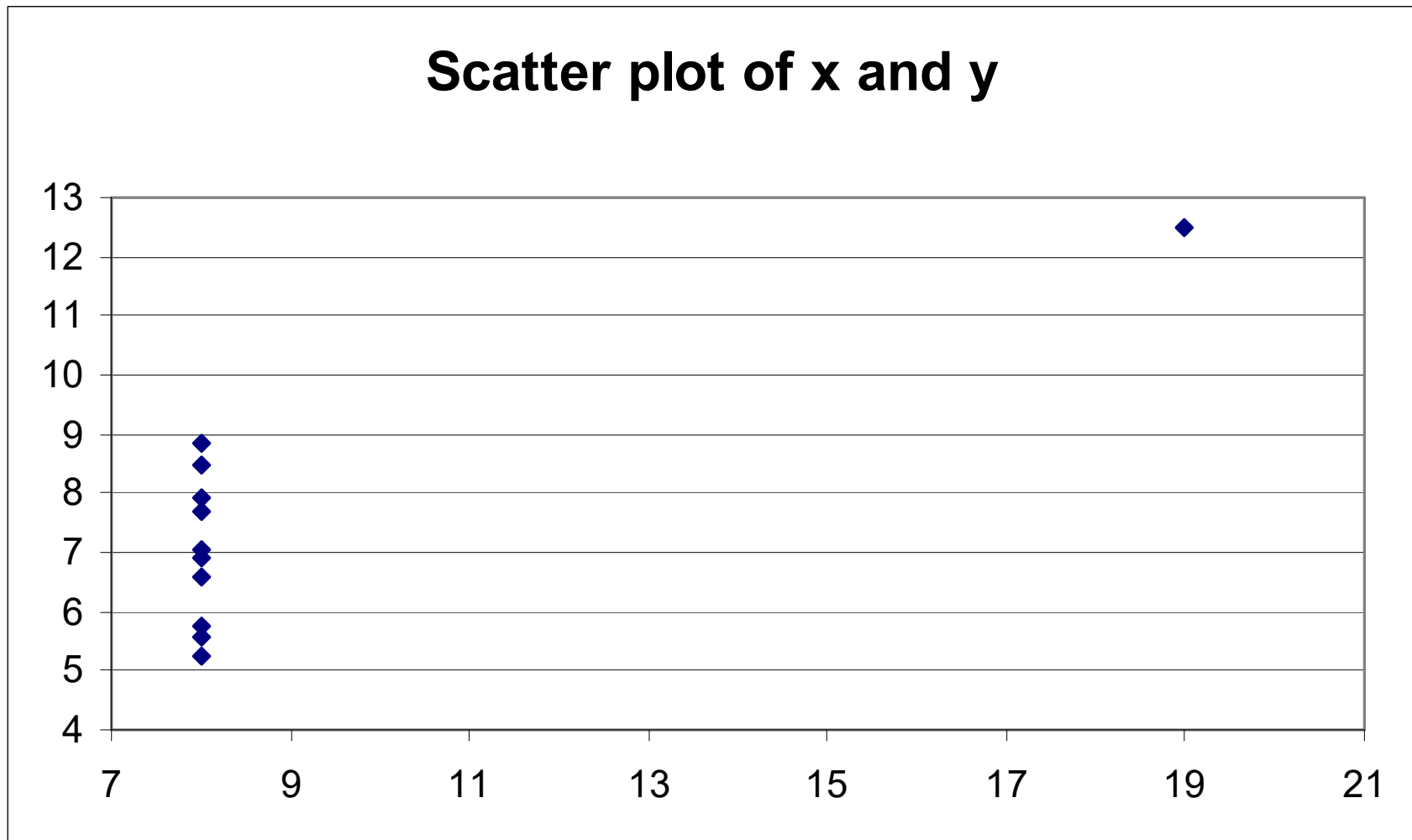
Source	DF	SS	MS	F	P
Regression	1	27.490	27.490	18.00	0.002
Residual Error	9	13.742	1.527		
Total	10	41.232			

Unusual Observations

Obs	x	y	Fit	SE Fit	Res.	St.R.
8	19.0	12.500	12.500	1.236	0.0	* X

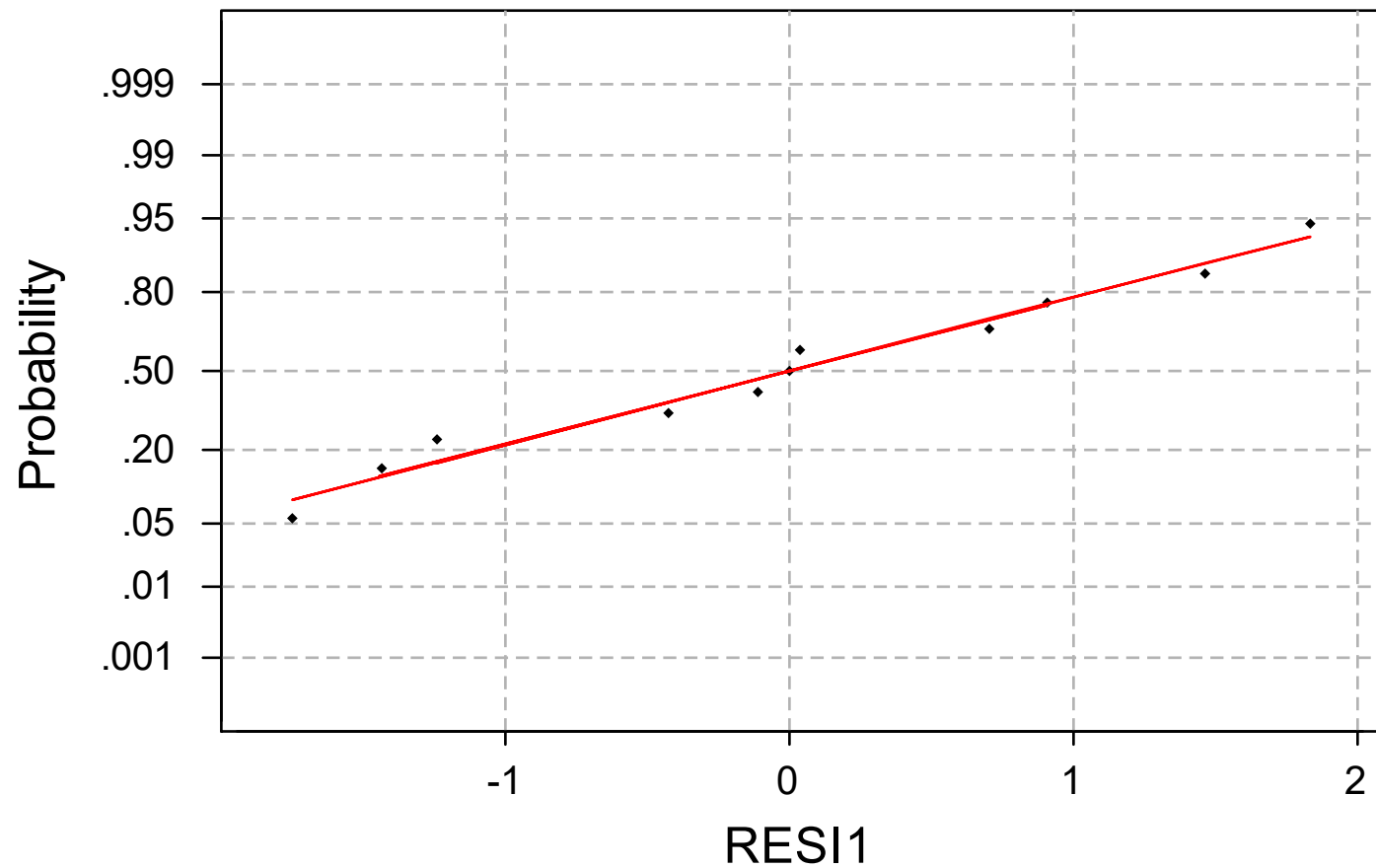
X denotes an observation whose X value gives it large influence.

Example (set 4)
with an influential observation



Example (set 4) with an influential observation

Normal Probability Plot



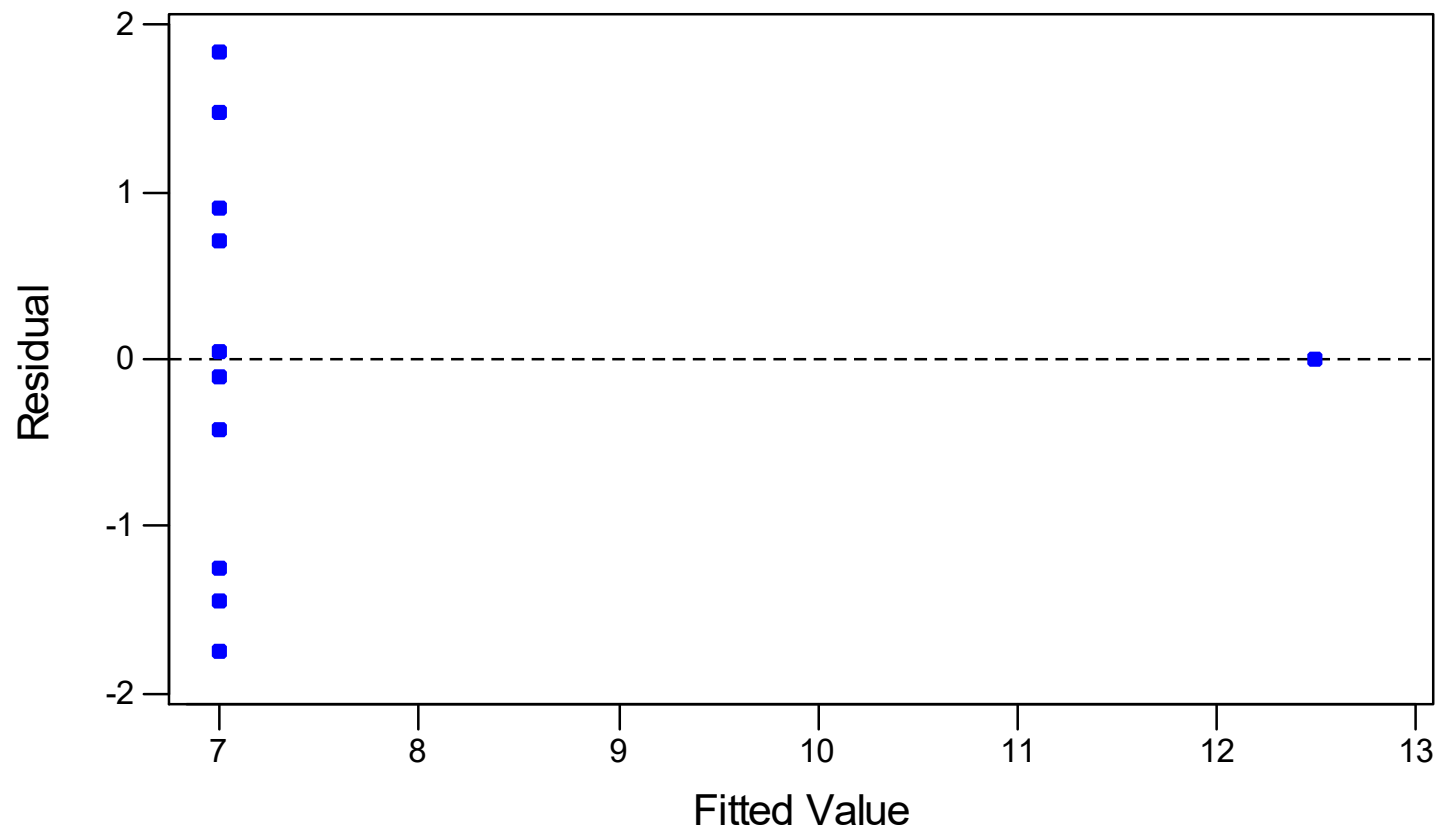
Average: -0.0000000
StDev: 1.17228
N: 11

W-test for Normality
R: 0.9872
P-Value (approx): > 0.1000

Example (set 4) with an influential observation

Residuals Versus the Fitted Values

(response is y)



Example (set 4) with an influential observation

- What if we remove the influential observation from the data set?